Agent-GhibliLock: Deterministic Wavelet-Locked Cross-Modal Style Transfer and ISO/IEC 19566-5 C2PA-Inspired Provenance Compilation for Multimodal Generative Pipelines

Aniket Agrawal
Cloud Engineer, Google Cloud
Google, Bangalore, India
aniketagrawal@google.com

***Abstract*—Enterprise multimodal workflows increasingly chain vision-language models (VLMs) with image-conditioned or adapter-guided latent diffusion image and video synthesizers. When an upstream VLM translates a source photograph into natural-language style instructions and a downstream diffusion model synthesizes restyled pixels (Pixels $\rightarrow$ Prose $\rightarrow$ Pixels), three coupled failure modes arise, which we term *The Captioning Conundrum*: (1) *high-frequency structural phase drift*, where even under macro-registered image or adapter conditioning, fine-grained corporate emblems, typographic glyphs, and perspective rails are degraded across the low-entropy text and $8\times$ latent VAE bottlenecks; (2) *temporal luminance flicker in video Groups of Pictures (GOPs)*, where frame-wise stochastic denoising induces severe exposure variance while naive temporal smoothing blurs moving edges; and (3) *cryptographic provenance severance and host-signal interference*, where container-level Coalition for Content Provenance and Authenticity (C2PA) manifests are stripped during cross-modal regeneration and social-platform re-encoding while blind additive spread-spectrum watermarks suffer host-signal interference. We present Agent-GhibliLock, a deterministic, CPU-executable post-processing compiler that reconciles high-frequency structural preservation, target palette alignment, temporal GOP stability, and dual-layer provenance attestation. First, Agent-GhibliLock decomposes CIELAB luminance (L^*) via an orthonormal K-level 2D Discrete Haar Wavelet Transform (DWT_2), supporting both *global* subband locking and *spatially masked region-of-interest (ROI)* detail locking (weighted by 5×5 Sobel directional coherence so macro-registered foreground emblems and rails are phase-locked while non-ROI background regions retain smooth painterly gouache brushwork), and proves that on dyadic grids ($2^K \mid H, W$) arbitrary spatial detail masks $M_k(u,v) \in [0,1]$ preserve exact pre-clamping spatial mean luminance $\mu(\widehat{L}^*_{\text{raw}}) = \mu(\widetilde{L}^*_{\text{cand}})$ and exact post-clamping mean under bounded interior drift (Theorem 1). Second, it aligns full 3D (L^*, a^*, b^*) color distributions via Iterative Sliced-Wasserstein Optimal Transport (SWOT) over $D = 3M+3 = 27$ orthonormal axes ($I_3 \cup \{Q_m\}_{m=1}^{M} \subset O(3)$) with deterministic lexicographical tie-breaking, explicit post-Step-A luminance mean re-anchoring, and a decoupled $(\tau_\gamma, \tau_\alpha)$ closed-loop controller satisfying an exact deterministic root-mean-square (RMS) Wasserstein perturbation bound (Theorem 2). Third, for GOP video synthesis, it suppresses inter-frame flicker via translation-compensated, innovation-gated wavelet LL_K exponential moving average (EMA) filtering while committing frame hashes to a binary Merkle tree root. Fourth, it binds hard and soft provenance by injecting a compact C2PA-inspired ISO/IEC 19566-5 JUMBF binary manifest box (`caBX` PNG chunk with IEEE 802.3 CRC-32 and keyed HMAC-SHA256) alongside a blind Quantization-Index-Modulation (QIM) watermark in mid-band 8×8 2D-DCT AC luminance coefficients supporting both $M_{\text{coord}} = 4$ ($R = \lfloor B/16 \rfloor$) and $M_{\text{coord}} = 5$ odd-repetition ($R = \lfloor 5B/64 \rfloor$) clipped-soft/hard majority voting, keyed dither K_{secret}, analytical AC ℓ_1 rounding bounds ($< \Delta_{\text{QIM}}/4$), and blind coset tamper localization (Theorem 3). Across a 64×64 stratified $N = 600$ controlled parametric candidate-degradation benchmark (evaluating 9 paradigms including a simulated ControlNet $8\times$-VAE bottleneck adapter baseline, $K \in \{1,2,3\}$ wavelet depth ablations, and disaggregated 4×25 adversarial provenance regimes), live multi-model Vertex AI Gemini evaluations (`gemini-2.5-flash`, `gemini-2.5-pro`, `gemini-3-flash-preview`), and multi-resolution natural photographic case studies (`grace_hopper.jpg` at 64^2–512^2, `Minduka_Present_Blue_Pack.png`, and a 24-image natural portrait/product cohort), Agent-GhibliLock achieves $\mathrm{SSIM}_{\text{edge}} = 0.9967$ (0.9977 ROI-masked; 0.9929–0.9966 global and 0.9845–0.9908 ROI-masked on real photographs), non-circular $\mathrm{SSIM}_{\text{full}} = 0.9752$, Sobel gradient correlation $\rho_{\text{grad}} = 0.8943$, post-export $\widehat{W}_2^{\text{out}} = 1.9614$ (held-out $\widehat{SW}_2^{\text{eval}} = 2.0741$), 99.9% GOP flicker reduction on rigid-shift exposure-flicker sequences, and 100.0% tamper detection and blind watermark bit recovery under container stripping, ± 2 LSB noise, and lossy JPEG $Q = 85$ recompression.**

***Index Terms*—Cross-Modal Style Transfer, Discrete Haar Wavelet Transform, Sliced-Wasserstein Optimal Transport, C2PA JUMBF Provenance, Quantization Index Modulation, 2D-DCT Watermarking, Multimodal Agentic Pipelines.**

I. Introduction

Modern enterprise creative and media workflows frequently orchestrate multi-stage generative pipelines in which a vision-language model (VLM) inspects an input asset I_{src} and emits structured scene descriptions, stylistic prompts, and conditioning priors for a downstream image-conditioned or adapter-guided latent diffusion synthesizer [1]–[4]. Unlike unconstrained text-to-image generation from scratch (where camera pose and macro-layout are synthesized freely), enterprise restyling pipelines separate coarse spatial conditioning from stylistic translation: image-to-image initialization [3], spatial ControlNet adapters [4], or homography/optical-flow registra-

tion maintain coarse macro-alignment, while VLM-generated prose and scalar conditioning priors (Pixels $\rightarrow$ Prose $\rightarrow$ Pixels) govern artistic palette, brushwork intensity, and structural retention (e.g., converting live-action photography into hand-painted Studio Ghibli cel-shaded gouache aesthetics). Throughout this work, *macro-registered* means that coarse semantic regions and foreground contours of the candidate I_{cand} remain pixel-grid aligned with I_{src} within the $2^K \times 2^K$ dyadic Haar tile support (4 px for our default $K = 2$, targeting sub-latent phase restoration within the $8\times$ VAE bottleneck; for higher-resolution synthesis where spatial adapters drift beyond 4 px, scaling K as $\lfloor \log_2(\min(H, W)/16) \rfloor$ suppresses luminance-plane double-edges, while optical-flow pre-warping of I_{cand} to I_{src} prior to Stage 1 is required to align candidate chrominance boundaries $(\widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$, as analyzed in Section IV). Even under macro-spatial conditioning, routing stylistic and structural control through a discrete token bottleneck and a downsampled latent autoencoder introduces three coupled failure modes that we term **The Captioning Conundrum**:

1) **High-Frequency Structural Phase Drift:** Natural language tokens have low spatial entropy (10^2–10^3 tokens $\approx$ 1–2 KB) compared to the high-frequency phase spectrum of an image ($H \times W \times 3$ bytes). Even when paired with iterative VLM-as-a-Judge retries, image-conditioned SDEdit [3], or latent $8\times$-downsampled ControlNet edge/tile adapters [4], fine-scale corporate logos (such as a 2 px geometric diamond shirt emblem), typographic glyphs, eyeglasses, collar insignia, and diagonal architectural rails suffer subband phase drift, spatial blurring across the $8\times$ VAE bottleneck, or local stroke hallucination [5].
2) **Temporal Luminance Flicker in Video GOPs:** When frame-by-frame or chunked diffusion synthesizers generate stylized video Groups of Pictures (GOPs), independent stochastic denoising trajectories induce severe frame-to-frame low-frequency luminance and exposure jumps ($\sigma^2_{\text{flicker}} > 340$). While naive pixel-space temporal exponential moving average (EMA) filtering blurs moving foreground edges, classical blind video temporal consistency methods such as Bonneel *et al.* [6] suppress flicker by solving dense optical-flow-warped gradient-domain screened Poisson equations at significant per-frame computational cost.
3) **Cryptographic Provenance Severance and Host-Signal Interference:** Regenerating pixels through an upstream diffusion decoder severs container-level Coalition for Content Provenance and Authenticity (C2PA) manifests [7]–[9], and social media platforms routinely strip ancillary metadata chunks (`caBX`, EXIF, XMP) upon upload. In the pixel domain, classical blind additive spread-spectrum DCT watermarks [20] suffer from severe Host-Signal Interference (HSI) when decoded without the unwatermarked host image. While modern deep neural watermarking frameworks such as TrustMark [10], SynthID-Image [19], and localized segmentation-based neural watermarks [24] overcome HSI and geometric distortions via learned encoder-decoder networks, they require neural network inference and do not provide analytical sub-Gaussian bit-error bounds or zero-GPU-dependency 8×8 block-level algebraic coset tamper localization, motivating host-rejecting lattice Quantization Index Modulation (QIM) [21].

Existing neural style transfer methods [11] and wavelet whitening-and-coloring transforms (WCT^2) [12] preserve spatial contours by operating directly on encoder-decoder feature maps, but rely on second-order Gaussian moment matching that fails to capture multi-modal cel-shaded gouache color distributions and provides neither closed-loop Wasserstein guarantees nor cryptographic C2PA/watermark provenance. Conversely, classical 1D/3D optimal transport (OT) color grading [16]–[18] aligns empirical color histograms, but applying raw pointwise rank-sorting to unconstrained diffusion candidates introduces contour staircasing and destroys source high-frequency edge fidelity unless paired with post-transport gradient-domain Poisson reconstruction [16], multi-scale regularization [17], [18], or continuous $\mathcal{O}(HW)$ spatial filterbanks such as Burt–Adelson Laplacian pyramids [22] and Guided Image Filtering [23] (which we benchmark directly in Section III alongside screened-Poisson temporal deflickering [6]; unlike smooth overlapping filterbanks, our discontinuous 2×2 orthonormal Haar basis provides exact block-wise zero-sum detail synthesis under arbitrary spatial ROI masks, boundary-free Parseval orthogonality, and exact alignment with the downstream 8×8 DCT watermark lattice).

To bridge this gap in a single deterministic pass without post-hoc Poisson PDE solvers or GPU watermark networks, we introduce **Agent-GhibliLock**, a zero-GPU-dependency cross-modal compilation engine that operates as a verifiable post-processing filterbank and provenance compiler on macro-registered candidate assets. To ensure complete experimental clarity regarding live API execution versus offline benchmark reproducibility, our evaluation explicitly separates: (i) **live Stage-0 VLM visual-to-text decomposition calls** executed against production Google Cloud Vertex AI endpoints (`gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` in `project="<gcp-project-id>"`, `location="global"`), where the VLM's emitted `structural_retention_prior` and style prompt parameterize both controlled and end-to-end SDXL + ControlNet candidate generation, (ii) a **controlled 64×64 parametric candidate-degradation harness** over $N = 600$ stratified trials that isolates deterministic compiler convergence from stochastic diffusion seed variance, and (iii) a **Real Natural Photographic Evaluation** across multi-resolution portraits and product photography (`grace_hopper.jpg` at 64×64 to 512×512, `Minduka_Present_Blue_Pack.png` at 128×128, and a 24-image natural photographic validation suite at

128×128 and 512×512).

A. Key Contributions

1) **Global and Spatially Masked ROI Haar Wavelet Detail Locking with Exact Mean Conservation (Theorem 1):** We formulate an orthonormal K-level 2D Discrete Haar Wavelet Transform (DWT_2/IDWT_2) [13]–[15] on CIELAB luminance L^* that supports both global subband injection and spatially masked ROI detail locking ($\Gamma_{k,\tau_\gamma}(u,v) = \gamma_{\tau_\gamma} M_k(u,v)$ derived via 5×5 Sobel directional coherence and 2×2 block max-pooling). We prove algebraically that on dyadic grids ($2^K \mid H, W$), multiplying Haar detail subbands (LH_k, HL_k, HH_k) by any spatially varying mask $M_k(u,v) \in [0,1]$ preserves the exact pre-clamping spatial mean luminance $\mu(\widehat{L}^*_{\text{raw}}) = \mu(\widetilde{L}^*_{\text{cand}})$ because every 2×2 Haar detail synthesis basis vector has zero spatial sum, and restores exact post-clamping mean $\mu(\widehat{L}^*) = \mu(L^*_{\text{ref}})$ whenever the interior clamping drift satisfies $|\Delta_\mu| \cdot (HW/|\mathcal{I}^\circ|) \leq 1$.
2) **3D Sliced-Wasserstein Optimal Transport with Exact Orthonormal-Frame Perturbation Bound (Theorem 2):** We align 3D CIELAB (L^*, a^*, b^*) distributions via iterative 1D optimal transport over $D = 3M + 3 = 27$ orthonormal axes ($Q_0 = I_3$ plus $M = 8$ Haar-uniform $O(3)$ frames via positive-diagonal QR factorization) with stable lexicographical tie-breaking (`mergesort`), explicit post-Step-A luminance mean re-anchoring, and a decoupled $(\tau_\gamma, \tau_\alpha)$ closed-loop controller. Denoting by $\mu_{\widetilde{I}_{\text{cand}}}$, $\mu_{\widehat{I}}$, and $\mu_{Y_{\text{ref}}}$ the empirical 3D CIELAB pixel distributions of the Step-A color-transported candidate $\widetilde{I}_{\text{cand}}$, the Step-B wavelet-locked image $\widehat{I}$, and the target style palette Y_{ref}, respectively, and by $\|\cdot\|_F$ the Frobenius norm over the $H \times W$ luminance plane, we prove via the orthogonality identity $\sum_{j=1}^{3} Q^2_{m,1j} = 1$ ($\forall Q_m \in O(3)$) an exact deterministic bound on the root-mean-square (RMS) Sliced-Wasserstein perturbation $|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})| \leq (3HW)^{-1/2} \|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\|_F$ with zero Monte Carlo slack.
3) **Motion-Compensated and Motion-Gated Video GOP Wavelet EMA Deflickering:** For multi-frame video synthesis, we formulate a translation-compensated ($\mathcal{T}_{\Delta u_t, \Delta v_t} \overline{LL}^{(t-1)}_{K,\text{out}}$), innovation-gated temporal EMA filter on the coarse LL_K wavelet subband paired with deterministic scalar DC mean re-anchoring. Specifically, while the deterministic scalar DC mean re-anchoring step algebraically eliminates global frame-mean flicker ($\sigma^2_{\text{flicker,pre-clamp}} < 10^{-12}$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$), the motion-compensated, innovation-gated wavelet EMA filter in Eq. (18) is evaluated via local coarse-subband tile temporal variance (σ^2_{tile}, reduced by 84.2–91.0%) and foreground $\mathrm{SSIM}_{\text{edge}}$ (suppressing motion ghosting on deforming foregrounds by $+0.5273$ over ungated EMA) while binding frame digests into a C2PA-inspired GOP Merkle root.
4) **Dual-Layer C2PA-Inspired JUMBF (`caBX`) Container and Blind Mid-Band 2D-DCT QIM Watermark with Exact AC Rounding Bounds (Theorem 3):** We combine a compact ISO/IEC 19566-5 JUMBF `caBX` PNG chunk with a blind mid-band 8×8 2D-DCT AC QIM watermark supporting $M_{\text{coord}} \in \{4, 5\}$ ($R = \lfloor 5B/64 \rfloor = 5$ odd repetition voting at 64×64, $\gcd(5, 64) = 1$), optional keyed dither K_{secret}, analytical AC ℓ_1 rounding bounds ($\max_{\mathcal{M}_4} |\eta_b^{\text{export}}| \leq 3.3486 < 3.5 = \Delta_{\text{QIM}}/4$), and clipped-soft/hard majority-voted blind coset tamper localization requiring zero source registry.

II. System Architecture and Mathematical Formulation

A. Stage 0: Cross-Modal Visual-to-Text Decomposition and Perceptual CIELAB Bijection

Given an input RGB image $I_{\text{src}} \in \{0, \ldots, 255\}^{H \times W \times 3}$ and a target style reference palette $Y_{\text{ref}} \in \{0, \ldots, 255\}^{H_s \times W_s \times 3}$, Stage 0 invokes a multimodal VLM (`gemini-2.5-flash`, `gemini-2.5-pro`, or `gemini-3-flash-preview`) with deterministic decoding (`temperature = 0.0`, `max_output_tokens = 256`, and `thinking_budget = 0` on Flash models) using the structured JSON prompt schema:

```
"You are the Stage-0 Visual-to-Text
Decomposer (GhP) for Agent-GhibliLock.
Scene metadata: {scene_summary}.
Return compact JSON with keys:
{\"ghp_prompt\": \"<concise Studio
Ghibli gouache style prompt
preserving emblem & diagonal rail>\",
\"structural_retention_prior\": <float
in [0.05, 0.15]>}"
```

The VLM returns both the natural-language style prompt (`ghp_prompt`, logged in the C2PA provenance manifest and forwarded to upstream text-conditioned synthesizers in full end-to-end deployments) and a scalar structural retention prior $\hat{\eta}_{\text{VLM}} \in [0.05, 0.15]$ (`structural_retention_prior`). Prior to Stage 0 and Stage 1, if $H_s \times W_s \neq H \times W$, Y_{ref} is spatially resampled onto the $H \times W$ grid via nearest-neighbor resampling (or constructed directly at $H \times W$ from I_{src} via 7×7 box-smoothed 6-band cel-shaded warm golden-hour/Prussian-blue grading on portrait/product scenes so that $(a^*_{\text{ref}}, b^*_{\text{ref}})$ is spatially co-registered with the semantic regions of I_{src}). In our controlled reproducible benchmark harness, $\eta_{\text{struct}} = \hat{\eta}_{\text{VLM}}$ (or $\eta_{\text{struct}} \in [0.04, 0.20]$ across curriculum tiers C1–C5) directly parameterizes the macro-registered candidate generator $I_{\text{cand}} = \mathcal{G}_{\text{cand}}(I_{\text{src}}, Y_{\text{ref}}; \eta_{\text{struct}})$, which blends source high-frequency luminance with low-pass 5×5 box-filtered ($\mathcal{B}_5$, applied channel-wise with replicate boundary padding) and phase-perturbed (3×3 shift-jittered ξ_{phase}) style-shifted channels:

$$\begin{aligned} L^*_{\text{cand}} &= (1 - \eta_{\text{struct}})(\mathcal{B}_5 * L^*_{\text{src}} + \Delta L^*_{\text{drift}} + \xi_{\text{phase}}) + \eta_{\text{struct}} L^*_{\text{src}}, \\ (a^*_{\text{cand}}, b^*_{\text{cand}}) &= (1 - \eta_{\text{struct}})(\mathcal{B}_5 * (a^*_{\text{ref}}, b^*_{\text{ref}}) + \Delta \mathbf{c}_{\text{drift}}) + \eta_{\text{struct}} (a^*_{\text{src}}, b^*_{\text{src}}), \end{aligned} \tag{1}$$

where $\xi_{\text{phase}}(x,y) = (\mathcal{B}_3 * L^*_{\text{src,no-logo}}(\text{clip}(x + \delta_x, 0, H - 1), \text{clip}(y + \delta_y, 0, W - 1))) - \mathcal{B}_5 * L^*_{\text{src}}(x,y)$ models 3×3

box-blurred sub-latent phase jitter with integer spatial shift $(\delta_x, \delta_y) \in \{-1, 0, 1\}^2$ (default $(1, -1)$) and emblem stroke erasure $L^*_{\text{src,no-logo}}$, and $(\Delta L^*_{\text{drift}}, \Delta \mathbf{c}_{\text{drift}}) \in \mathbb{R} \times \mathbb{R}^2$ models uncalibrated upstream diffusion color drift ($\widehat{W}_2 \approx$ 14–26), instantiated in sRGB by blending the macro-blurred source at weight $\eta_{\text{macro}} = 0.24$ with a warm synthetic drift field $C_{\text{drift}}(x, y) = \text{clip}((175 - \lfloor 55y/H \rfloor, 135 + \lfloor 45x/W \rfloor, 95 + \lfloor 35y/H \rfloor) + \boldsymbol{\zeta}(x, y), 0, 255)$ with i.i.d. integer jitter $\boldsymbol{\zeta}(x, y) \sim \text{Unif}\{-5, \ldots, 5\}^3$.

To decouple perceptual lightness and structural edges from chromaticity, all three images $\{I_{\text{src}}, I_{\text{cand}}, Y_{\text{ref}}\}$ are mapped into the CIE 1976 (L^*, a^*, b^*) color space under the CIE Standard Illuminant D65 white point $(X_n, Y_n, Z_n) = (0.95047, 1.00000, 1.08883)$. For normalized sRGB channels $c \in [0, 1]$, inverse companding (IEC 61966-2-1) yields linear light:

$$c_{\text{lin}} = \begin{cases} c/12.92, & \text{if } c \le 0.04045, \\ ((c + 0.055)/1.055)^{2.4}, & \text{otherwise.} \end{cases} \tag{2}$$

Linear RGB is projected to CIE XYZ via the exact IEC 61966-2-1 D65 matrix $M_{\text{RGB}\to\text{XYZ}} = \left[\begin{smallmatrix} 0.4124564 & 0.3575761 & 0.1804375 \\ 0.2126729 & 0.7151522 & 0.0721750 \\ 0.0193339 & 0.1191920 & 0.9503041 \end{smallmatrix}\right]$ and transformed via the piecewise cubic root transfer function $f(t)$ with threshold $\delta = 6/29$:

$$f(t) = \begin{cases} t^{1/3}, & t > \delta^3, \\ \dfrac{t}{3\delta^2} + \dfrac{4}{29}, & \text{else,} \end{cases} \quad \begin{aligned} L^* &= 116 f(Y/Y_n) - 16, \\ a^* &= 500\,(f(X/X_n) - f(Y/Y_n)), \\ b^* &= 200\,(f(Y/Y_n) - f(Z/Z_n)). \end{aligned} \tag{3}$$

The inverse map LAB $\to$ sRGB applies the analytical inverse f^{-1}, multiplies by $M_{\text{XYZ}\to\text{RGB}} = M^{-1}_{\text{RGB}\to\text{XYZ}} = \left[\begin{smallmatrix} 3.2404542 & -1.5371385 & -0.4985314 \\ -0.9692660 & 1.8760108 & 0.0415560 \\ 0.0556434 & -0.2040259 & 1.0572252 \end{smallmatrix}\right]$, clips out-of-gamut linear values to $[0, 1]$, applies forward gamma companding, and rounds to uint8, achieving round-trip error ≤ 1 LSB across $\{0, \ldots, 255\}^3$.

B. Stage 1: Orthonormal Multi-Level 2D Haar Wavelet Decomposition and Structural Hashing

Let $L^* \in [0, 100]^{H \times W}$ denote a 2D luminance plane. When H or W is odd at any decomposition level, a single row or column is padded via edge replication prior to 2×2 analysis and cropped back to the exact target dimensions upon inverse synthesis.[1] For each 2×2 block $\mathbf{B}_{u,v} = [c_{00}, c_{01};\ c_{10}, c_{11}]$, the single-level orthonormal 2D Discrete Haar Wavelet Transform (DWT_2) [13], [14] computes:

$$\begin{bmatrix} LL(u,v) \\ LH(u,v) \\ HL(u,v) \\ HH(u,v) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \begin{bmatrix} c_{00} \\ c_{01} \\ c_{10} \\ c_{11} \end{bmatrix} = H_2 \mathbf{c}_{u,v}. \tag{4}$$

Because the 4×4 Haar matrix H_2 is symmetric and orthonormal ($H_2^T = H_2^{-1} = H_2$), the single-level inverse transform IDWT_2 is exact ($\mathbf{c}_{u,v} = H_2^T [LL, LH, HL, HH]^T$) and satisfies Parseval's identity $\|L^*\|_F^2 = \|LL_K\|_F^2 + \sum_{k=1}^{K} (\|LH_k\|_F^2 + \|HL_k\|_F^2 + \|HH_k\|_F^2)$ for any K-level recursive cascade on dyadic dimensions. Unlike smooth overlapping filterbanks (e.g., Burt–Adelson Laplacian pyramids [22] or Guided Image Filtering [23], evaluated in Section III), non-overlapping 2×2 Haar synthesis satisfies $\mathbf{1}_4^T H_2^T = [2, 0, 0, 0]$ on every 2×2 tile, guaranteeing exact spatial mean conservation under arbitrary spatially varying ROI masks $M_k(u, v) \in [0, 1]$ (Theorem 1) while aligning directly with the 8×8 Stage-3 DCT watermark grid.

From the Level-1 detail subbands (LH_1, HL_1, HH_1) of L^*_{src}, we define the half-resolution structural edge energy map $E_{1/2}(u, v)$ and its $2\times$ nearest-neighbor upsampled, scale-normalized full-resolution map $E_{\text{full}}(L^*) \in [0, 1]^{H \times W}$:

$$\begin{aligned} E_{1/2}(u,v) &= \sqrt{LH_1(u,v)^2 + HL_1(u,v)^2 + \tfrac{1}{2} HH_1(u,v)^2}, \\ E_{\text{full}}(x,y) &= \frac{E_{1/2}(\lfloor x/2 \rfloor, \lfloor y/2 \rfloor)}{\max\left(\max_{u,v} E_{1/2}(u,v),\ 10^{-8}\right)}. \end{aligned} \tag{5}$$

Thresholding $E_{\text{full}}(L^*_{\text{src}})$ at $q_{0.75} = \max(\text{Quantile}_{0.75}(E_{\text{full}}),\ 0.08)$ yields the binary structural edge/logo mask $M_{\text{HF}} \in \{0, 1\}^{H \times W}$. To bind the source geometry into the C2PA manifest, M_{HF} is serialized as a contiguous row-major (C-order) uint8 byte buffer (`0x00`/`0x01`) concatenated with an 8-byte big-endian dimension trailer $\text{pack}(\texttt{">II"}, H, W)$ and hashed via SHA-256:

$$h_{\text{struct}} = \text{SHA256}\left(\text{bytes}_{\text{C-uint8}}(M_{\text{HF}}) \parallel \text{pack}(\texttt{">II"}, H, W)\right). \tag{6}$$

a) Structural Verification Certificate and Non-Circular Spatial Metrics: During closed-loop compilation, we monitor the global Structural Similarity Index ($\text{SSIM}_{\text{edge}}$) [5] between $E_1 = E_{\text{full}}(L^*_{\text{src}})$ and $E_2 = E_{\text{full}}(L^*_{\text{out}})$ with $c_1 = 0.01^2, c_2 = 0.03^2$, where $(\mu_{E_1}, \mu_{E_2}, \sigma^2_{E_1}, \sigma^2_{E_2}, \sigma_{E_1 E_2})$ are evaluated in a single global pass over all $N = HW$ pixels using population normalization $(1/N)$:

$$\text{SSIM}_{\text{edge}}(L^*_{\text{src}}, L^*_{\text{out}}) = \frac{(2\mu_{E_1}\mu_{E_2} + c_1)(2\sigma_{E_1 E_2} + c_2)}{(\mu^2_{E_1} + \mu^2_{E_2} + c_1)(\sigma^2_{E_1} + \sigma^2_{E_2} + c_2)}. \tag{7}$$

Because $\text{SSIM}_{\text{edge}}$ operates on the scale-normalized Level-1 Haar detail energy map E_{full}, it serves as an internal *compiler verification certificate* confirming that source detail subbands were retained across sRGB gamut clipping and Stage-3 DCT QIM watermarking. However, because E_{full} is invariant to positive global contrast scaling (up to the 10^{-8} denominator stabilizer in Eq. 5) and local sign flips ($LH_1 \mapsto -LH_1$), relying solely on $\text{SSIM}_{\text{edge}}$ would be partially self-referential. We therefore pair $\text{SSIM}_{\text{edge}}$ in both the closed-loop stopping criterion and all benchmark evaluations with two non-circular spatial-domain metrics: (i) $\text{SSIM}_{\text{full}}$, computed directly over non-overlapping 4×4 spatial luminance patches ($x = L^*_{\text{src}}/100, y = L^*_{\text{out}}/100$, stride 4) without wavelets by evaluating the two-factor contrast-and-structure similarity $\text{clip}\left(\frac{2\sigma_x\sigma_y + c_2}{\sigma_x^2 + \sigma_y^2 + c_2} \cdot \max\left(0, \frac{\sigma_{xy} + c_3}{\sigma_x \sigma_y + c_3}\right), 0, 1\right)$ ($c_2 = 0.03^2, c_3 = c_2/2$) with population variance/covariance

[1] Exact Parseval energy conservation and zero-mean detail synthesis hold without boundary terms whenever $2^K \mid H$ and $2^K \mid W$ (e.g., 64×64, 128×128, 256×256, 512×512); for non-dyadic odd dimensions, edge replication introduces an $O(1/\min(H, W))$ boundary approximation.

normalization ($1/N_p$ with $N_p = 16$ pixels per patch) independently on each 4×4 patch and averaging uniformly across all $\lfloor H/4 \rfloor \lfloor W/4 \rfloor$ patches; and (ii) signed 3×3 Sobel gradient inner-product correlation $\rho_{\text{grad}} \in [-1, 1]$ using the standard unnormalized 3×3 Sobel kernels $K_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ and $K_y = K_x^T$ evaluated over the interior structural edge locus $M_{\text{HF}}^{\circ} = \{(x, y) \in \{1, \ldots, H-2\} \times \{1, \ldots, W-2\} \mid M_{\text{HF}}(x, y) = 1\}$ (excluding the 1-pixel image border where 3×3 Sobel filters require boundary padding):

$$\rho_{\text{grad}}(L^*_{\text{src}}, L^*_{\text{out}}) = \frac{\sum_{(x,y) \in M_{\text{HF}}^{\circ}} \langle \nabla_{\text{Sobel}} L^*_{\text{src}}(x, y),\ \nabla_{\text{Sobel}} L^*_{\text{out}}(x, y) \rangle}{\|\nabla_{\text{Sobel}} L^*_{\text{src}}\|_{2, M_{\text{HF}}^{\circ}} \cdot \|\nabla_{\text{Sobel}} L^*_{\text{out}}\|_{2, M_{\text{HF}}^{\circ}} + 10^{-12}}, \quad (8)$$

which reverses sign ($\rho_{\text{grad}} \to -1$) under edge polarity inversions and penalizes 4×4 block boundary discontinuities.

C. Stage 2: 3D Sliced-Wasserstein Optimal Transport and Wavelet-Locked Subband Fusion

a) Step A: 3D CIELAB Sliced-Wasserstein Optimal Transport (SWOT): Let $X = \{x_i\}_{i=0}^{N-1} \subset \mathbb{R}^3$ ($N = HW$) denote the flat row-major sequence of CIELAB pixels of I_{cand}, and let $Y = \{y_j\}_{j=0}^{N-1} \subset \mathbb{R}^3$ denote the target style palette Y_{ref} nearest-neighbor resampled to $H \times W$ when $H_s \times W_s \neq H \times W$ (throughout Eqs. 11, 13, and Theorem 1, $L^*_{\text{ref}} \in [0, 100]^{H \times W}$ denotes the luminance plane of this $H \times W$ resampled reference Y_{ref}). For any unit direction $\theta \in \mathbb{S}^2$, the 1D projected empirical Wasserstein-2 distance is $W_2(\theta_{\#}\mu_X, \theta_{\#}\mu_Y) = \left(\frac{1}{N} \sum_{r=1}^{N} (\langle x_{\pi_\theta(r)}, \theta \rangle - \langle y_{\sigma_\theta(r)}, \theta \rangle)^2 \right)^{1/2}$, where $\pi_\theta, \sigma_\theta : \{1, \ldots, N\} \to \{0, \ldots, N-1\}$ are stable sorting bijections mapping 1-based rank r to the 0-based flat row-major pixel index $i \in \{0, \ldots, N-1\}$ with ties broken deterministically in lexicographical order of i (`kind="mergesort"`). For any 1D reference array $Z = (z_0, \ldots, z_{N-1}) \in \mathbb{R}^N$ with stably sorted order statistics $z_{\sigma_Z(1)} \leq \cdots \leq z_{\sigma_Z(N)}$, we define the 1D empirical quantile lookup operator $\mathcal{Q}_Z(\text{rank}(U))$ on $U \in \mathbb{R}^N$ by mapping the r-th stably sorted entry $u_{\pi_U(r)}$ to $z_{\sigma_Z(r)}$ for $r \in \{1, \ldots, N\}$.

To match the true quadratic Sliced-Wasserstein-2 metric [17], [18], we define both the in-sample training objective $\widehat{W}_2$ over the $D = 3M + 3 = 27$ orthonormal axes $\Theta_{\text{train}} = I_3 \cup \{Q_m\}_{m=1}^{M}$ ($M = 8$ Haar-uniform orthogonal frames $Q_m \in O(3)$ generated via positive-diagonal QR factorization $Z_m = Q_m R_m$, $Q_m \leftarrow Q_m \text{diag}(\text{sign}(\text{diag}(R_m)))$ from `seed=42`)[2] and the out-of-sample evaluation metric $\widehat{SW}_2^{\text{eval}}$ over $M_{\text{eval}} = 64$ independent uniform unit-sphere directions (`seed=999`) as root-mean-square (RMS) estimators:

$$\widehat{W}_2(\mu_X, \mu_Y) = \left(\frac{1}{D} \sum_{r=1}^{D} W_2^2(\theta_{r\#}\mu_X, \theta_{r\#}\mu_Y) \right)^{1/2},$$
$$\widehat{SW}_2^{\text{eval}}(\mu_X, \mu_Y) = \left(\frac{1}{M_{\text{eval}}} \sum_{s=1}^{M_{\text{eval}}} W_2^2(\tilde{\theta}_{s\#}\mu_X, \tilde{\theta}_{s\#}\mu_Y) \right)^{1/2}. \quad (9)$$

Starting from $X^{(-1)} = X$, Step A initializes $X^{(0)}$ via coordinate-wise 1D quantile transport along $Q_0 = I_3$ ($m = 0$ in Eq. 10) with damped step size $\alpha_{\tau_\alpha} \in [0.85, 0.97]$, and iterates across $m = 1, \ldots, M$ orthonormal bases $Q_m \in O(3)$:

$$X^{(m)} = X^{(m-1)} + \alpha_{\tau_\alpha} \sum_{d=1}^{3} \left(\mathcal{Q}_{Y q_{m,d}}(\text{rank}(X^{(m-1)} q_{m,d})) - X^{(m-1)} q_{m,d} \right) q_{m,d}^T, \quad (10)$$

applies a sequential half-step polish ($0.5\alpha_{\tau_\alpha}$) over the ordered tuple (I_3, Q_{M-1}, Q_M) (re-executing Eq. 10 sequentially for $m \in (0, M-1, M)$ with step size $0.5\alpha_{\tau_\alpha}$ prior to Eq. 11, since sequential rank-projection updates do not commute), and enforces an explicit post-Step-A luminance mean re-anchoring shift:

$$\widetilde{L}^*_{\text{cand}}(x, y) \leftarrow \widetilde{L}^*_{\text{cand}}(x, y) + \left(\mu(L^*_{\text{ref}}) - \mu(\widetilde{L}^*_{\text{cand}}) \right), \quad (11)$$

guaranteeing that $\mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$ holds identically entering Step B (where $\widetilde{L}^*_{\text{cand}} \in \mathbb{R}^{H \times W}$ remains unclamped prior to Step B synthesis, $\widehat{W}_2^{\text{StepA}} = \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})$ is evaluated immediately after Eq. 11, and by the orthogonal-frame identity $\frac{1}{D} \sum_{r=1}^{D} (\theta_{r,L^*})^2 = \frac{1}{3}$ of Theorem 2, the scalar shift $\Delta\mu_{\text{StepA}} = \mu(L^*_{\text{ref}}) - \mu_{\text{pre}}(\widetilde{L}^*_{\text{cand}})$ perturbs pre-shift $\widehat{W}_2$ by at most $|\Delta\mu_{\text{StepA}}|/\sqrt{3}$).

b) Step B: Global vs. Spatially Masked ROI Wavelet Detail Locking: Applying pointwise rank-order sorting in Step A aligns the 3D color distribution $\widetilde{I}_{\text{cand}} = (\widetilde{L}^*_{\text{cand}}, \widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$ but does not restore high-frequency source geometry and can introduce subtle 4×4 tile-boundary quantile steps in $LL_K(\widetilde{L}^*_{\text{cand}})$. Furthermore, in natural photographs with noisy backgrounds or skies, locking 100% of high-frequency wavelet details globally transfers photographic sensor grain into background regions that should exhibit smooth Studio Ghibli gouache brushwork.

Agent-GhibliLock therefore supports both **Global Wavelet Locking** ($M_{\text{ROI}}(x, y) \equiv 1$) and **Spatially Masked ROI Wavelet Locking**, where a smooth spatial weighting mask $M_{\text{ROI}}(x, y) \in [0, 1]^{H \times W}$ is constructed from $E_{\text{full}}(L^*_{\text{src}})$ weighted by 5×5 Sobel directional coherence (using the unnormalized 3×3 Sobel kernels K_x, K_y for $(G_x, G_y) = \nabla_{\text{Sobel}} L^*_{\text{src}}$ with replicate `mode="edge"` boundary padding for ∇_{Sobel}, $\mathcal{B}_5$, and $\mathcal{B}_3$):

$$\mathcal{C}_{5\times5}(x, y) = \frac{\sqrt{(\mathcal{B}_5 * G_x)^2 + (\mathcal{B}_5 * G_y)^2}}{\mathcal{B}_5 * \sqrt{G_x^2 + G_y^2} + 10^{-6}},$$
$$W_{\text{edge}}(x, y) = E_{\text{full}}(L^*_{\text{src}})(x, y) \cdot \mathcal{C}_{5\times5}(x, y)^3. \quad (12)$$

Coherent structural contours, logos, and rails satisfy $\mathcal{C}_{5\times5} \geq 0.35$ and undergo grayscale morphological max-dilation over a

[2] We sample from the full orthogonal group $O(3)$ rather than $SO(3)$ because 1D Wasserstein-2 distance is reflection-invariant: $W_2((-\theta)_{\#}\mu_X, (-\theta)_{\#}\mu_Y) = W_2(\theta_{\#}\mu_X, \theta_{\#}\mu_Y)$.

5×5 square (ℓ_∞ radius 2 px) structuring element followed by 3×3 normalized box feathering $\mathcal{B}_3$ and scale normalization via $M_{\text{ROI}}(x,y) = \text{clip}\Big(\frac{\mathcal{B}_3 * \text{Dilate}_{5\times5}(W_{\text{edge}}\cdot\mathbf{1}\{\mathcal{C}_{5\times5}\geq0.35\})}{\max(\max_{x,y} W_{\text{edge}}(x,y), 10^{-8})}, 0, 1\Big) \in [0,1]$, whereas isolated background sensor grain ($\mathcal{C}_{5\times5} < 0.35$) is zeroed prior to dilation and receives exact weight 0. At each wavelet level $k \in \{1,\dots,K\}$, M_{ROI} is downsampled to subband dimensions $(H/2^k)\times(W/2^k)$ via 2×2 block max-pooling to form $M_k(u,v)\in[0,1]$, ensuring every 2×2 Haar block overlapping the ROI receives full locking weight $\Gamma_{k,\tau_\gamma}(u,v) = \gamma_{\tau_\gamma} M_k(u,v)$. Step B blends source detail subbands $D_k(L^*_{\text{src}})$ with the smooth pre-sort candidate detail subbands $D_k(L^*_{\text{cand}})$ (eliminating Step A rank-sort staircasing):

$$\begin{aligned}\widehat{LL}_K &= \mathcal{S}_{3\times3}\Big((1-\eta_{\tau_\alpha})LL_K(\widetilde{L}^*_{\text{cand}}) + \eta_{\tau_\alpha}\mathcal{Q}_{LL_K(L^*_{\text{ref}})}\Big(LL_K(\widetilde{L}^*_{\text{cand}})\Big)\Big),\\ \widehat{D}_k &= \Gamma_{k,\tau_\gamma}\odot D_k(L^*_{\text{src}}) + (1-\Gamma_{k,\tau_\gamma})\odot D_k(L^*_{\text{cand}}),\end{aligned} \tag{13}$$

for $D \in \{LH, HL, HH\}$ and $k \in \{1,\dots,K\}$, where $\mathcal{Q}_{LL_K(L^*_{\text{ref}})}$ is the exact tie-broken rank-order quantile bijection onto the $(H/2^K)\times(W/2^K)$ subband $LL_K(L^*_{\text{ref}})$ (which is a bijection of the empirical multiset preserving $\mu(\mathcal{Q}_{LL_K(L^*_{\text{ref}})}(Z)) = \mu(LL_K(L^*_{\text{ref}})) = 2^K\mu(L^*_{\text{ref}}))$ and $\mathcal{S}_{3\times3}(Z) = 0.65Z + 0.35(\mathcal{B}_3 * Z) + 0.35(\mu(Z) - \mu(\mathcal{B}_3 * Z))$ is a mean-preserving 3×3 coarse-subband smoothing operator.

c) Decoupled $(\tau_\gamma,\tau_\alpha)$ Closed-Loop Controller: Synthesizing $\widehat{L}^*_{\text{raw}} = \text{IDWT}_2(\widehat{LL}_K, \{\widehat{D}_k\}_{k=1}^K)$ can inflate global luminance variance when L^*_{src} has extreme contrast, which would increase $\widehat{W}_2^{\text{StepB}}$. Crucially, if a single coupled index τ simultaneously increased γ_{τ_γ} whenever $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, injecting *more* high-frequency source detail energy would move $\widehat{W}_2^{\text{StepB}}$ in the wrong direction. Agent-GhibliLock therefore initializes 0-based controller state indices $(\tau_\gamma,\tau_\alpha) \in \{0,1,2,3\}^2$ at $(0,0)$ and executes at most $\tau_{\max} = 4$ closed-loop passes (indexed by 1-based pass counter $\tau\in\{1,2,3,4\}$, with default thresholds $\tau_{\text{struct}} = 0.88$, $\tau_{\text{grad}} = 0.88$, and $\tau_{\text{color}} = 4.5$) over $\gamma_{\tau_\gamma}\in\{0.78,0.88,0.94,0.98\}$, $\alpha_{\tau_\alpha}\in\{0.85,0.90,0.94,0.97\}$, $\eta_{\tau_\alpha}\in\{0.00,0.45,0.75,0.95\}$, and variance-matching weight $\nu_{\tau_\alpha}\in\{0.00,0.72,0.90,0.98\}$. Within each loop pass τ, Step A is executed (or reused from cache whenever τ_α is unchanged from the previous pass, as well as across paired Global/ROI-Masked evaluations sharing $\tau_\alpha = 0$ at $\tau = 1$) and Step B evaluates $(\text{SSIM}_{\text{edge}}, \rho_{\text{grad}}, \widehat{W}_2^{\text{StepB}})$ directly on the CIELAB state $\widehat{I} = (\widehat{L}^*, \widetilde{a}^*_{\text{cand}}, \widetilde{b}^*_{\text{cand}})$; once the stopping criterion is satisfied (or $\tau = 4$ is reached), LAB $\to$ sRGB gamut clipping and Stage-3 2D-DCT QIM watermarking execute once outside the loop:

1) Prior to clamping, if $\tau_\alpha > 0$ and $\sigma(\widehat{L}^*_{\text{raw}}) > \sigma(\widetilde{L}^*_{\text{cand}})$, apply the affine variance-regularization map $\widehat{L}^*_{\text{raw}} \leftarrow \mu(\widetilde{L}^*_{\text{cand}}) + \kappa_{\tau_\alpha}(\widehat{L}^*_{\text{raw}} - \mu(\widehat{L}^*_{\text{raw}}))$ with $\kappa_{\tau_\alpha} = (1-\nu_{\tau_\alpha}) + \nu_{\tau_\alpha}\sigma(\widetilde{L}^*_{\text{cand}})/\sigma(\widehat{L}^*_{\text{raw}})$. Because Step 3 increments τ_α only when $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}} = 4.5$—and empirically $\widehat{W}_2^{\text{StepB}} \approx 1.7$–$2.3 \ll 4.5$ across standard trials—both η_{τ_α} in Eq. (13) and Step-1 variance regularization (ν_{τ_α}) remain inactive ($\eta_0 = 0, \nu_0 = 0, \kappa_0 = 1$) on pass $\tau = 1$ (and on subsequent passes triggered solely by $\text{SSIM}_{\text{edge}} < \tau_{\text{struct}}$ or $\rho_{\text{grad}} < \tau_{\text{grad}}$), acting specifically as fallback color/variance regularizers when extreme source contrast pushes $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$. This map preserves $\mu(\widehat{L}^*_{\text{raw}}) = \mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$ identically, and because positive affine scaling around the mean cancels analytically in E_{full} (Eq. 5) and ρ_{grad} (Eq. 8) up to the 10^{-8} and 10^{-12} denominator stabilizers prior to boundary clamping, it contracts $\widehat{W}_2^{\text{StepB}}$ without degrading $\text{SSIM}_{\text{edge}}$ or ρ_{grad}.
2) Clamp $\widehat{L}^*_{\text{clip}} = \text{clip}(\widehat{L}^*_{\text{raw}}, 0, 100)$ and apply interior DC re-anchoring over unsaturated pixels $\mathcal{I}^\circ = \{(x,y) : \widehat{L}^*_{\text{clip}}(x,y) \in (1,99)\}$ via $\widehat{L}^*(x,y) = \text{clip}(\widehat{L}^*_{\text{clip}}(x,y) + \delta_{\text{DC}}\mathbf{1}_{(x,y)\in\mathcal{I}^\circ}, 0, 100)$, where $\delta_{\text{DC}} = \text{sign}(\Delta_\mu)\min(|\Delta_\mu|HW/|\mathcal{I}^\circ|, 1.0)$ if $\mathcal{I}^\circ \neq \emptyset$ and $\delta_{\text{DC}} = 0$ if $\mathcal{I}^\circ = \emptyset$, so $\mu(\widehat{L}^*) = \mu(\widetilde{L}^*_{\text{cand}}) = \mu(L^*_{\text{ref}})$ holds identically whenever the interior drift condition $|\Delta_\mu|\cdot(HW/|\mathcal{I}^\circ|) \leq 1$ of Theorem 1 is met.
3) Evaluate $\text{SSIM}_{\text{edge}}(L^*_{\text{src}}, \widehat{L}^*)$, $\rho_{\text{grad}}(L^*_{\text{src}}, \widehat{L}^*)$, and $\widehat{W}_2^{\text{StepB}}(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}})$: if $\text{SSIM}_{\text{edge}} < \tau_{\text{struct}}$ or $\rho_{\text{grad}} < \tau_{\text{grad}}$, increment $\tau_\gamma \leftarrow \min(3, \tau_\gamma + 1)$ (and increment $\tau_\alpha \leftarrow \min(3, \tau_\alpha + 1)$ only if $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, holding τ_α fixed and reusing the cached Step-A $\widetilde{I}_{\text{cand}}$ whenever $\widehat{W}_2^{\text{StepB}} \leq \tau_{\text{color}}$); else if $\widehat{W}_2^{\text{StepB}} > \tau_{\text{color}}$, hold τ_γ fixed and increment only $\tau_\alpha \leftarrow \min(3, \tau_\alpha + 1)$; else terminate the loop.

Theorem 1 (Exact Spatial Mean Luminance Conservation under Spatially Masked Haar Detail Locking)**.** *Let $L^*_{src}, \widetilde{L}^*_{cand}, L^*_{ref} \in \mathbb{R}^{H\times W}$ have dyadic dimensions ($2^K \mid H, W$), where $\widetilde{L}^*_{cand}$ has been mean-anchored via Eq. (11) such that $\mu(\widetilde{L}^*_{cand}) = \mu(L^*_{ref})$. For any K-level Haar decomposition and any spatially varying detail mask $M_k(u,v) \in [0,1]^{(H/2^k)\times(W/2^k)}$, the pre-clamping wavelet-locked luminance $\widehat{L}^*_{raw} = \text{IDWT}_2(\widehat{LL}_K, \{\widehat{LH}_k, \widehat{HL}_k, \widehat{HH}_k\}_{k=1}^K)$ (both before and after the mean-preserving Step-1 affine variance regularization around $\mu(\widetilde{L}^*_{cand})$) satisfies:*

$$\mu(\widehat{L}^*_{raw}) = \frac{1}{HW}\sum_{x=0}^{H-1}\sum_{y=0}^{W-1}\widehat{L}^*_{raw}(x,y) = 2^{-K}\mu(\widehat{LL}_K) = \mu(\widetilde{L}^*_{cand}) = \mu(L^*_{ref}). \tag{14}$$

Moreover, let $\widehat{L}^*_{clip} = \text{clip}(\widehat{L}^*_{raw}, 0, 100)$ and let $\mathcal{I}^\circ = \{(x,y) : \widehat{L}^*_{clip}(x,y) \in (1,99)\}$ denote the interior unsaturated pixel locus. If $\mathcal{I}^\circ \neq \emptyset$ and the clamping drift $\Delta_\mu = \mu(L^*_{ref}) - \mu(\widehat{L}^*_{clip})$ satisfies $|\Delta_\mu|\cdot(HW/|\mathcal{I}^\circ|) \leq 1$, the piecewise interior DC restoration map $\widehat{L}^*(x,y) = \text{clip}\Big(\widehat{L}^*_{clip}(x,y) + \frac{HW}{|\mathcal{I}^\circ|}\Delta_\mu\mathbf{1}_{(x,y)\in\mathcal{I}^\circ}, 0, 100\Big)$ restores $\mu(\widehat{L}^*) = \mu(L^*_{ref})$ identically without boundary re-saturation.[3]

Proof. At any single inverse Haar synthesis level $k \in \{1,\dots,K\}$ (where $LL_K = \widehat{LL}_K$ and $LL_0 = \widehat{L}^*_{\text{raw}}$), each 2×2 block $\mathbf{c}_{u,v} \in \mathbb{R}^4$ of LL_{k-1} is reconstructed via

[3]Empirically, across all $N = 600$ benchmark trials and the $n = 100$ high-contrast stress suite, the interior condition $\mathcal{I}^\circ \neq \emptyset$ and $|\Delta_\mu|\cdot(HW/|\mathcal{I}^\circ|) \leq 1$ is satisfied on 100.0% (600/600 and 100/100) of trials, with maximum observed $|\Delta_\mu|\cdot(HW/|\mathcal{I}^\circ|) = 0.0842 \ll 1.0$.

$\mathbf{c}_{u,v} = H_2^T[LL_k(u,v), \widehat{LH}_k(u,v), \widehat{HL}_k(u,v), \widehat{HH}_k(u,v)]^T$. Multiplying on the left by $\mathbf{1}_4^T = [1,1,1,1]$ yields:

$$\mathbf{1}_4^T \mathbf{c}_{u,v} = (\mathbf{1}_4^T H_2^T) \begin{bmatrix} LL_k(u,v) \\ \widehat{LH}_k(u,v) \\ \widehat{HL}_k(u,v) \\ \widehat{HH}_k(u,v) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} LL_k(u,v) \\ \widehat{LH}_k(u,v) \\ \widehat{HL}_k(u,v) \\ \widehat{HH}_k(u,v) \end{bmatrix} = 2\,LL_k(u,v), \quad (15)$$

regardless of the values of $(\widehat{LH}_k(u,v), \widehat{HL}_k(u,v), \widehat{HH}_k(u,v)) = \Gamma_{k,\tau_\gamma}(u,v) D_k(L^*_{\text{src}})(u,v) + (1 - \Gamma_{k,\tau_\gamma}(u,v)) D_k(L^*_{\text{cand}})(u,v)$. Summing over all (u,v) at level k gives $\sum_{u',v'} LL_{k-1}(u',v') = 2\sum_{u,v} LL_k(u,v)$, and inducting from $k = K$ down to $k = 1$ yields $\sum_{x,y} \widehat{L}^*_{\text{raw}}(x,y) = 2^K \sum_{u,v} \widehat{LL}_K(u,v)$. Because the rank-order bijection $\mathcal{Q}_{LL_K(L^*_{\text{ref}})}$ is a permutation of $LL_K(L^*_{\text{ref}})$ on equal $(H/2^K) \times (W/2^K)$ grids and $\mathcal{S}_{3\times3}$ explicitly re-centers its output mean, $\mu(\widehat{LL}_K) = (1-\eta_{\tau_\alpha}) 2^K \mu(\widetilde{L}^*_{\text{cand}}) + \eta_{\tau_\alpha} 2^K \mu(L^*_{\text{ref}}) = 2^K \mu(L^*_{\text{ref}})$. Dividing by HW (and noting that Step-1 affine scaling around $\mu(\widetilde{L}^*_{\text{raw}}) = \mu(\widetilde{L}^*_{\text{cand}})$ leaves the mean invariant) yields Eq. (14). Finally, when $|\Delta_\mu|(HW/|\mathcal{I}^\circ|) \le 1$, the $\min(\cdot, 1.0)$ cap in Step 2 is inactive and every $(x,y) \in \mathcal{I}^\circ$ (where $\widehat{L}^*_{\text{clip}}(x,y) \in (1,99)$) satisfies $\widehat{L}^*_{\text{clip}}(x,y) + \frac{HW}{|\mathcal{I}^\circ|}\Delta_\mu \in (0,100) \subset [0,100]$, so the outer clip to $[0,100]$ is inactive and summing over $\{0,\dots,H-1\} \times \{0,\dots,W-1\}$ yields $\mu(\widehat{L}^*) = \mu(\widehat{L}^*_{\text{clip}}) + \Delta_\mu = \mu(L^*_{\text{ref}})$. □

Theorem 2 (Exact Deterministic RMS Sliced-Wasserstein Perturbation Bound)**.** *Let $\widetilde{I}_{cand} = (\widetilde{L}^*_{cand}, \widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$ and $\widehat{I} = (\widehat{L}^*, \widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$ share identical transported chromaticity channels $(\widetilde{a}^*_{cand}, \widetilde{b}^*_{cand})$. Over any collection of $M+1$ complete orthonormal frames $\{Q_0, Q_1, \dots, Q_M\} \subset O(3)$ ($D = 3(M+1)$ unit projection axes), the root-mean-square Sliced-Wasserstein-2 distance $\widehat{W}_2$ (Eq. 9) satisfies the exact deterministic bound:*

$$\left|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})\right| \le \widehat{W}_2(\mu_{\widehat{I}}, \mu_{\widetilde{I}_{\text{cand}}}) \le \frac{1}{\sqrt{3HW}} \left\|\widehat{L}^* - \widetilde{L}^*_{cand}\right\|_F. \quad (16)$$

Proof. Because $\widehat{W}_2$ is an L_2 root-mean-square norm of 1D Wasserstein-2 distances across $r \in \{1,\dots,D\}$, the reverse triangle inequality implies $|\widehat{W}_2(\mu_{\widehat{I}}, \mu_{Y_{\text{ref}}}) - \widehat{W}_2(\mu_{\widetilde{I}_{\text{cand}}}, \mu_{Y_{\text{ref}}})| \le \widehat{W}_2(\mu_{\widehat{I}}, \mu_{\widetilde{I}_{\text{cand}}})$. For each axis $\theta_r = (\theta_{r,L^*}, \theta_{r,a^*}, \theta_{r,b^*}) \in \mathbb{S}^2$, the Hardy-Littlewood-Pólya rearrangement inequality implies that sorting is a non-expansive operator in $\ell_2(\mathbb{R}^N)$ ($N = HW$):

$$W_2^2(\theta_{r\#}\mu_{\widehat{I}}, \theta_{r\#}\mu_{\widetilde{I}_{\text{cand}}}) \le \frac{1}{N}\sum_{i=1}^{N} \langle \widehat{I}_i - \widetilde{I}_{\text{cand},i}, \theta_r\rangle^2 = \frac{(\theta_{r,L^*})^2}{HW} \left\|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\right\|_F^2, \quad (17)$$

since $\widehat{a}^* - \widetilde{a}^*_{\text{cand}} = 0$ and $\widehat{b}^* - \widetilde{b}^*_{\text{cand}} = 0$. Averaging over the $D = 3(M+1)$ axes partitioned into $M+1$ orthogonal matrices $Q_m \in O(3)$, the three columns of each Q_m form an orthonormal basis of $\mathbb{R}^3$, so the first row of every Q_m has unit Euclidean norm: $\sum_{j=1}^3 (Q_{m,1j})^2 = 1$. Consequently, $\frac{1}{D}\sum_{r=1}^{D} (\theta_{r,L^*})^2 = \frac{1}{3(M+1)}\sum_{m=0}^{M} 1 = \frac{1}{3}$ holds *deterministically and identically* for all $M \ge 0$. Taking square roots yields Eq. (16). □

D. Temporal Video GOP Extension: Motion-Compensated and Motion-Gated Wavelet EMA

For an 8-frame Group of Pictures (GOP) $\{I^{(t)}_{\text{src}}, I^{(t)}_{\text{cand}}\}_{t=1}^T$, independent diffusion synthesis causes both global frame-to-frame luminance flicker $\sigma^2_{\text{flicker}} = \text{Var}_t(\overline{L}^{*(t)} - \overline{L}^{*(t-1)})$ (where $\overline{L}^{*(t)} = \mu(L^{*(t)})$ is the frame mean luminance and Var_t is the unbiased sample variance with denominator $T-2$ over the $T-1$ inter-frame differences) and local coarse-subband tile temporal variance $\sigma^2_{\text{tile}} = \frac{1}{(H/2^K)(W/2^K)}\sum_{u,v} \text{Var}_t(2^{-K}\overline{LL}^{(t)}_{K,\text{out}}(u,v) - 2^{-K}\overline{LL}^{(t-1)}_{K,\text{out}}(u,v))$ over the compiled output subbands $\overline{LL}^{(t)}_{K,\text{out}}$, whereas unwarped temporal EMA filtering ghosts across moving objects. At frame $t=1$, Agent-GhibliLock executes the single-frame closed-loop controller to convergence and initializes $\overline{LL}^{(1)}_{K,\text{out}} = \widehat{LL}^{(1)}_K$ from Eq. (13), locking the converged controller state $(\tau_\gamma, \tau_\alpha)$ and variance-regularization weight ν_{τ_α} (with $\kappa_0 = 1$ when $\tau_\alpha = 0$) across the GOP. For each subsequent frame $t \in \{2,\dots,T\}$, Step-A 3D SWOT (Eq. 10) executes on $I^{(t)}_{\text{cand}}$ using the converged index τ_α from frame 1, and Step-B subband fusion produces $\widehat{LL}^{(t)}_K$ and $\{\widehat{D}^{(t)}_k\}_{k=1}^K$ once per frame t outside the inner controller loop. Agent-GhibliLock then estimates the integer-tile translation $(\Delta u_t, \Delta v_t) \in \{-4,\dots,4\}^2$ on the $(H/2^K) \times (W/2^K)$ subband grid (corresponding to 2^K-pixel dyadic shifts; optional full-resolution integer-pixel pre-alignment can be applied prior to DWT_2) between the source coarse subbands $LL^{(t-1)}_{K,\text{src}} = LL_K(L^{*(t-1)}_{\text{src}})$ and $LL^{(t)}_{K,\text{src}} = LL_K(L^{*(t)}_{\text{src}})$ by minimizing the zero-mean cross-correlation objective $\text{MSE}(\Delta u, \Delta v) = \|(LL^{(t)}_{K,\text{src}} - \mu(LL^{(t)}_{K,\text{src}})) - \mathcal{T}_{\Delta u,\Delta v}(LL^{(t-1)}_{K,\text{src}} - \mu(LL^{(t-1)}_{K,\text{src}}))\|^2_F$, where the translation operator is defined with explicit index-shift sign convention $\mathcal{T}_{\Delta u,\Delta v}(Z)(u,v) = Z(\text{clip}(u - \Delta u, 0, H/2^K - 1), \text{clip}(v - \Delta v, 0, W/2^K - 1))$ (or cyclic index shift $(u - \Delta u) \bmod (H/2^K)$ on periodic scenes, breaking ties lexicographically in $(|\Delta u| + |\Delta v|, \Delta u, \Delta v)$, and accepting non-zero shifts only when MSE drops by $\ge 35\%$ relative to $(\Delta u, \Delta v) = (0,0)$). It then computes the scale-normalized motion innovation $\Delta^{(t)}_{\text{mot}}(u,v) = 2^{-K}|LL^{(t)}_{K,\text{src}}(u,v) - \mathcal{T}_{\Delta u_t,\Delta v_t} LL^{(t-1)}_{K,\text{src}}(u,v)|$ (evaluated directly on uncentered source subbands for constant-exposure source video GOPs, or on mean-centered source subbands $LL^{(t)}_{K,\text{src}} - \mu(LL^{(t)}_{K,\text{src}})$ when natural source illumination varies across the GOP) and applies the continuous (C^0) hinged excess-innovation gated coarse-subband update:

$$\beta_t(u,v) = \beta \exp\left(-\frac{\left(\max(0,\, \Delta^{(t)}_{\text{mot}}(u,v) - \sigma_{\text{mot}})\right)^2}{2\sigma^2_{\text{mot}}}\right),$$
$$\overline{LL}^{(t)}_{K,\text{out}} = \beta_t \odot \mathcal{T}_{\Delta u_t,\Delta v_t} \overline{LL}^{(t-1)}_{K,\text{out}} + (1 - \beta_t) \odot \widehat{LL}^{(t)}_K, \quad (18)$$

with $\beta = 0.78, \sigma_{\text{mot}} = 35.0$, followed by global mean re-anchoring $\overline{LL}^{(t)}_{K,\text{out}} \leftarrow \overline{LL}^{(t)}_{K,\text{out}} + (2^K \mu(L^*_{\text{ref}}) - \mu(\overline{LL}^{(t)}_{K,\text{out}}))$ prior to $\text{IDWT}_2(\overline{LL}^{(t)}_{K,\text{out}}, \{\widehat{D}^{(t)}_k\}_{k=1}^K)$ synthesis, after which

Step-1 affine variance regularization (using ν_{τ_α} from frame 1 and recomputing $\sigma(\widetilde{L}^{*(t)}_{\text{cand}})/\sigma(\widehat{L}^{*(t)}_{\text{raw}})$ when $\tau_\alpha > 0$, or $\kappa_0 = 1$ when $\tau_\alpha = 0$) and Step-2 interior DC clamping/restoration (δ_{DC}) execute on every frame $t \geq 1$. Crucially, for metric attribution, by Theorem 1 the deterministic scalar DC mean re-anchoring shift alone algebraically guarantees $\overline{L}^{*(t)} = \mu(L^*_{\text{ref}})$ across all frames $t \in \{1, \ldots, T\}$ ($\sigma^2_{\text{flicker,pre-clamp}} < 10^{-12}$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$, even if $\beta = 0$, while applying a negligible uniform shift to static background tiles when $\beta_t(u, v)$ varies spatially); therefore, local coarse-subband tile temporal variance σ^2_{tile} and foreground $\text{SSIM}_{\text{edge}}$ are the primary non-trivial metrics evaluating the temporal wavelet EMA and motion-innovation gate $\beta_t(u, v)$ in Eq. (18), whereas $\sigma^2_{\text{flicker}}$ verifies global DC exposure locking. Each compiled output frame $I^{(t)}_{\text{out}}$ is serialized in canonical row-major C-order uint8 bytes to form 32-byte binary SHA-256 digests $h_t = \text{SHA256}(\text{bytes}_{\text{C-uint8}}(I^{(t)}_{\text{out}}))$, which are folded pairwise by hashing concatenated raw 32-byte digests $\text{SHA256}(h_{2k-1} \,\|\, h_{2k})$ (duplicating the final digest at any level with an odd node count when T is not a power of 2) into the binary Merkle root R_{GOP}.

E. Stage 3: Dual-Layer Provenance—Mid-Band 2D-DCT QIM Watermark and C2PA-Inspired JUMBF (`caBX`)

a) Mid-Band 2D-DCT Quantization-Index-Modulation (QIM) Watermark: To survive metadata chunk stripping, Stage 3 embeds a $N_w = 64$-bit cryptographic payload $\mathbf{w} \in \{0, 1\}^{64}$ (unpacked MSB-first from the first 8 raw digest bytes $\text{SHA256}(h_{\text{struct}} \,\|\, \texttt{model_ids})\texttt{[:8]}$, where h_{struct} is concatenated as its 64-byte lowercase ASCII hex digest and `model_ids` is UTF-8 encoded in the canonical format `"<vlm_id>|<diffusion_id>"`) into the floating-point ITU-R BT.601 luminance channel $Y = 0.299R + 0.587G + 0.114B \in [0, 255]$ partitioned into $B = \lfloor H/8 \rfloor \lfloor W/8 \rfloor$ non-overlapping 8×8 blocks. Each shifted block $\mathbf{B}_b - 128$ is transformed via the orthonormal 8×8 Type-II DCT matrix C_8: $\mathbf{F}_b = C_8(\mathbf{B}_b - 128)C_8^T$. Crucially, we never modify the DC $(0, 0)$ coefficient, restricting embedding strictly to mid-band AC coordinates under two supported layouts:

- **Standard 4-Mode AC Layout ($M_{\text{coord}} = 4$):** $\mathcal{M}_4 = \{(2, 3), (3, 2), (1, 3), (3, 1)\}$, yielding $R = \lfloor 4B/64 \rfloor = \lfloor B/16 \rfloor$ votes per bit ($R = 4$ at 64×64, $R = 16$ at 128×128, $R = 64$ at 256×256, $R = 256$ at 512×512).
- **Odd-Repetition 5-Mode AC Layout ($M_{\text{coord}} = 5$):** $\mathcal{M}_5 = \mathcal{M}_4 \cup \{(2, 2)\}$, yielding $R = \lfloor 5B/64 \rfloor$ votes per bit, which produces strictly odd majority votes whenever $B/64$ is odd (such as $R = 5$ at 64×64 with $\gcd(5, 64) = 1$, eliminating even-vote ties at 64×64 and dispersing bit assignments across both rows and columns).

For block $b \in \{0, \ldots, B-1\}$ and mode $m \in \{0, \ldots, M_{\text{coord}} - 1\}$, bit index i is assigned via $\phi(b, m) = (5b + m) \bmod 64$ under $M_{\text{coord}} = 5$ (where $\gcd(5, 64) = 1$ guarantees exact uniform multiplicity $R = 5B/64$ across all 64 bits whenever $64 \mid B$), whereas for $M_{\text{coord}} = 4$ we apply the row-shifted mapping $\phi_{\text{row}}(b, m) = (4b + 4r_b + m) \bmod 64$ with $r_b = \lfloor b/\lfloor W/8 \rfloor \rfloor$ whenever $\gcd(4\lfloor W/8 \rfloor, 64) = 64$ (i.e., $16 \mid \lfloor W/8 \rfloor$, such as 128×128, 256×256, and 512×512; for $\lfloor W/8 \rfloor = 8$ at 64×64, the unshifted $\phi(b, m) = (4b + m) \bmod 64$ already shifts consecutive block rows by 32 $(\text{mod } 64)$ while preserving exact uniform multiplicity $R = 4$). Bit w_i is embedded with step size $\Delta_{\text{QIM}} = 14.0$ and optional keyed pseudo-random dither $d_{b,m}(K_{\text{secret}}) = \Delta_{\text{QIM}} \cdot (\texttt{uint32_be}(\text{HMAC-SHA256}(K_{\text{secret}}, \text{pack}(\texttt{">II"}, b, m))\texttt{[:4]})/2^{32}) \in [0, \Delta_{\text{QIM}})$ ($d_{b,m} \equiv 0$ in unkeyed public audit mode) [21], where $\lfloor x + 1/2 \rfloor$ in Eq. (19) denotes the standard floor function implementing round-half-up quantization on the QIM lattice:

$$\begin{aligned} \widetilde{F}_b(u_m, v_m) &= \left\lfloor \frac{F_b(u_m, v_m) - \delta_{b,m}(w_i)}{\Delta_{\text{QIM}}} + \frac{1}{2} \right\rfloor \Delta_{\text{QIM}} + \delta_{b,m}(w_i), \\ \delta_{b,m}(w_i) &= d_{b,m}(K_{\text{secret}}) + \frac{\Delta_{\text{QIM}}}{2} w_i. \end{aligned} \tag{19}$$

Denoting the shifted coset lattices by $\Lambda_{w_i}(b, m) = \Delta_{\text{QIM}}\mathbb{Z} + \delta_{b,m}(w_i)$ for $w_i \in \{0, 1\}$ and the minimum Euclidean coset distance by $\text{dist}(z, \Lambda_{w_i}(b, m)) = \min_{k \in \mathbb{Z}} |z - (k\Delta_{\text{QIM}} + \delta_{b,m}(w_i))|$, the spatial luminance increment $\Delta Y = (\text{IDCT}_8(\widetilde{\mathbf{F}}) + 128) - Y = \text{IDCT}_8(\widetilde{\mathbf{F}} - \mathbf{F})$ (applied block-wise via $C_8^T(\widetilde{\mathbf{F}}_b - \mathbf{F}_b)C_8$ on each 8×8 block $b \in \{0, \ldots, B - 1\}$) is added equally across (R, G, B) and rounded/clamped to $\{0, \ldots, 255\}$. During extraction, each block/mode $(b, m) \in \mathcal{B}_i$ carrying bit i contributes either a binary hard vote $v^{\text{hard}}_{b,m} = \mathbf{1}\{\text{dist}(F^{\text{rx}}_b(u_m, v_m), \Lambda_1(b, m)) < \text{dist}(F^{\text{rx}}_b(u_m, v_m), \Lambda_0(b, m))\} \in \{0, 1\}$ or a **clipped soft coset vote** $v_{b,m} = \text{clip}\Big(1 - \frac{\text{dist}(F^{\text{rx}}_b(u_m, v_m), \Lambda_1(b,m))}{\Delta_{\text{QIM}}/2}, 0, 1\Big) \in [0, 1]$ with decision rule $\hat{w}_i = \mathbf{1}\Big\{\sum_{(b,m) \in \mathcal{B}_i} v_{b,m} > |\mathcal{B}_i|/2\Big\}$ (where $|\mathcal{B}_i| = R$ for all $i \in \{0, \ldots, 63\}$ whenever $64 \mid M_{\text{coord}}B$, or by restricting embedding/extraction to the first $64\lfloor B/64 \rfloor$ blocks when $64 \nmid M_{\text{coord}}B$), ensuring no single tampered block can contribute a vote outside $[0, 1]$.

Theorem 3 (QIM Coset Minimum-Distance Margin with Exact Mid-Band AC Rounding Bounds and Blind Tamper Localization)**.** *Let $\widetilde{F}_b(u_m, v_m) \in \Lambda_{w_i}(b, m)$ be a quantized mid-band AC DCT coefficient ($(u_m, v_m) \in \mathcal{M}_4$ or $\mathcal{M}_5$, assuming $64 \mid M_{coord}B$ so $|\mathcal{B}_i| = R$ for all i) before* uint8 *export. Upon adding ΔY equally to integer pre-watermark channels $(R_0, G_0, B_0) \in \{0, \ldots, 255\}^3$, rounding to $\mathbb{Z}$, clamping to $[0, 255]$, and applying an external channel perturbation, the received DCT coefficient is $F^{rx}_b(u_m, v_m) = \widetilde{F}_b(u_m, v_m) + \eta^{export}_b(u_m, v_m) + \eta^{chan}_b(u_m, v_m)$.*

1) ***Exact Mid-Band AC ℓ_1 Rounding Bounds and Cross-Channel Export Self-Noise:*** *Since $\text{dist}(\Lambda_0(b, m), \Lambda_1(b, m)) = \Delta_{QIM}/2$, a single coefficient decodes error-free whenever $|\eta^{export}_b + \eta^{chan}_b| < \Delta_{QIM}/4 = 3.5$. For interior blocks whose pre-watermark channels lie in $[\lceil \|\Delta Y\|_\infty \rceil, 255 - \lceil \|\Delta Y\|_\infty \rceil]$ (where $\|\Delta Y\|_\infty \leq \frac{\Delta_{QIM}}{2} \max_{x,y} \sum_m |c_{u_m}(x) c_{v_m}(y)| \leq 5.54$ LSB for $\mathcal{M}_4$ and ≤ 7.04 LSB for $\mathcal{M}_5$ [or via factored norms $\leq \frac{\Delta_{QIM}}{2} \sum_m \|c_{u_m}\|_\infty \|c_{v_m}\|_\infty \leq 6.54$ LSB for $\mathcal{M}_4$ and ≤ 8.04 LSB for $\mathcal{M}_5$, since $\|c_1\|_\infty = \|c_3\|_\infty = \frac{1}{2}\cos(\pi/16) \approx 0.4904$ at $x = 2$ and*

$\|c_2\|_\infty = \frac{1}{2}\cos(\pi/8) \approx 0.4619$*] across all superposed modes, so* $c_0 + \Delta Y \in [0, 255]$*), adding* ΔY *equally to integer* $R_0, G_0, B_0 \in \mathbb{Z}$ *yields* $c_0 + \Delta Y$ *with identical fractional parts* $\{\Delta Y\}$ *across all three channels, so nearest-integer rounding (under either round-half-up or IEEE 754 round-half-to-even tie-breaking) produces quantization errors* $|\epsilon_R|, |\epsilon_G|, |\epsilon_B| \le 0.5$ *satisfying* $|\epsilon_Y| = |0.299\epsilon_R + 0.587\epsilon_G + 0.114\epsilon_B| \le 0.5$*. While the unwatermarked DC* $(0,0)$ *mode has* ℓ_1 *norm* $\|c_0\|_1 = \sqrt{8}$ *(worst-case bound* 4.0*), the 1D Type-II DCT AC basis vectors* c_1, c_2, c_3 *satisfy* $\|c_1\|_1 = \|c_3\|_1 = \frac{1}{2}\csc(\pi/16) \approx 2.5629$ *and* $\|c_2\|_1 = \sqrt{4+2\sqrt{2}} \approx 2.6131$*, yielding exact worst-case interior rounding bounds* $B_{\max}$ *strictly below the QIM Voronoi radius* $\Delta_{QIM}/4 = 3.5$*:*

$$\max_{(u_m,v_m)\in\mathcal{M}_4} |\eta_b^{export}(u_m, v_m)| = \tfrac{1}{2}\|c_2\|_1\|c_3\|_1 \approx 3.3486 < 3.5,$$
$$\max_{(u_m,v_m)\in\mathcal{M}_5} |\eta_b^{export}(u_m, v_m)| = \tfrac{1}{2}\|c_2\|_1^2 = 2+\sqrt{2} \approx 3.4142 < 3.5. \tag{20}$$

Hence interior uint8 *rounding alone can never flip a single watermarked AC coefficient (*$p_e^{export} = 0$*). Under uniform quantization noise* $\epsilon_Y \sim \mathrm{Unif}[-0.5, 0.5]$*, orthonormality of* C_8 *yields theoretical interior rounding variance* $\sigma^2_{round} = 1/12 \approx 0.0833$ *(*RMS ≈ 0.2887*; empirical* $\sigma^2_{round+clamp} = 0.0821 \ll 12.25$*).*

2) ***R-Fold Repetition Voting:*** *Because* $M_{coord} \in \{4, 5\}$ *(with* $M_{coord} < 64$*) and* $\phi(b, m)$ *increments by* $1 \pmod{64}$ *across* $m \in \{0, \ldots, M_{coord} - 1\}$*, the* R *votes for any bit* i *reside in* $|\mathcal{B}_i| = R$ *spatially disjoint* 8×8 *blocks (with keyed dither* $d_{b,m}(K_{secret})$ *decorrelating quantization residuals across blocks). Under the standard spatially white (mutually independent) bounded quantization-noise model for* $\epsilon_Y(x, y) \in [-0.5, 0.5]$ *paired with zero-mean sub-Gaussian external channel noise* $\eta_b^{chan}(u_m, v_m)$ *of variance proxy* σ^2_η*, the total perturbation* $\eta^{tot}_{b,m} = \eta_b^{export}(u_m, v_m) + \eta_b^{chan}(u_m, v_m)$ *has sub-Gaussian variance proxy* σ^2_{eff} *bounded by* $\sigma^2_{round,Hoeff} = 0.25 + \sigma^2_\eta$ *(or approximated empirically by* $\sigma^2_{round+clamp} + \sigma^2_\eta$*), yielding* $p_e \le 2\exp(-\Delta^2_{QIM}/(32\sigma^2_{eff}))$*; alternatively, without assuming intra-block pixel independence of* $\epsilon_Y(x, y)$*, combining the exact deterministic export bound* $B_{\max} \in \{3.3486, 3.4142\} < \Delta_{QIM}/4$ *from Eq. (20) directly with the sub-Gaussian tail bound on* η_b^{chan} *over the reduced Voronoi margin* $\Delta_{QIM}/4 - B_{\max}$ *guarantees* $p_e \le 2\exp(-(\Delta_{QIM}/4 - B_{\max})^2/(2\sigma^2_\eta))$*. For odd* R*, hard majority voting across disjoint blocks yields* $P_{bit} = \sum_{k=(R+1)/2}^{R} \binom{R}{k} p_e^k (1-p_e)^{R-k}$ *(satisfying* $P_{bit} \le 10p_e^3$ *for all* $p_e \in [0, 1]$ *at* $R = 5$*), whereas even* R *includes the tie-breaking term* $\frac{1}{2}\binom{R}{R/2} p_e^{R/2}(1-p_e)^{R/2}$*.*

3) ***Combinatorial Guarantee for Blind Coset Tamper Localization:*** *Let* $\mathcal{B}_i = \{b : \exists m, \phi(b, m) = i\}$ *(*$|\mathcal{B}_i| = R$*) denote the* R *distinct* 8×8 *blocks carrying bit* i*, and let* $\mathcal{T} \subset \{0, \ldots, B-1\}$ *denote the set of spatially tampered blocks with tamper fraction* $\alpha_i = |\mathcal{T} \cap \mathcal{B}_i|/R$*. Under hard majority voting, whenever untampered blocks* $b \notin \mathcal{T}$ *satisfy* $|\eta_b^{export} + \eta_b^{chan}| < \Delta_{QIM}/4$ *and* $|\mathcal{T} \cap \mathcal{B}_i| \le \lfloor (R-1)/2 \rfloor$ *(*$\alpha_i < 1/2$*) for every bit* $i \in \{0, \ldots, 63\}$*,* $\hat{\mathbf{w}} = \mathbf{w}$ *is recovered with zero errors regardless of tamper amplitude; under clipped-soft voting* $v_{b,m} \in [0, 1]$*, exact recovery holds whenever untampered blocks satisfy* $|\eta_b^{export} + \eta_b^{chan}| \le \delta_{clean} < \frac{\Delta_{QIM}}{2}\frac{1/2-\alpha_i}{1-\alpha_i}$ *(which reduces to* $\delta_{clean} < \frac{\Delta_{QIM}}{2(R+1)}$ *at the worst-case* $\alpha_i = \frac{R-1}{2R}$ *for odd* R*, strictly above the conservative lower bound* $\frac{\Delta_{QIM}}{4R}$*). Once* $\hat{\mathbf{w}}$ *is recovered, blind block-level tamper localization flags block* b *via* $\max_m \mathrm{dist}(F_b^{rx}(u_m, v_m), \Lambda_{\hat{w}_{\phi(b,m)}}(b, m)) > \Delta_{QIM}/4$ *in lossless/PNG mode, or when* ≥ 3 *modes exceed* $0.4\Delta_{QIM}$ *in JPEG-resilient mode (automatically selected when the input container is JPEG or the high-frequency DCT noise floor* $\hat{\sigma}_{HF} = \mathrm{median}_{b\in\{0,\ldots,B-1\}}\left(\frac{1}{|\mathcal{H}_{HF}|}\sum_{(u,v)\in\mathcal{H}_{HF}} |F_b^{rx}(u, v)|^2\right)^{1/2}$ *over unwatermarked diagonal high-frequency AC modes* $\mathcal{H}_{HF} = \{(u, v) \in \{0, \ldots, 7\}^2 \setminus (\{(0, 0)\} \cup \mathcal{M}_{coord}) : u + v \ge 6\}$ *exceeds* $0.18\Delta_{QIM}$*).*

Proof. Since $\eta_b^{\text{export}}(u, v) = \sum_{x=0}^{7}\sum_{y=0}^{7} c_u(x)c_v(y)\epsilon_Y(x, y)$, Hölder's inequality with $\|\epsilon_Y\|_\infty \le 0.5$ (achieved when $\epsilon_Y(x, y) = 0.5\,\mathrm{sign}(c_u(x)c_v(y))$) gives $\max|\eta_b^{\text{export}}(u, v)| = \frac{1}{2}\|c_u\|_1\|c_v\|_1$. Direct evaluation of $\|c_k\|_1 = \sum_{x=0}^{7}\frac{1}{2}|\cos(\frac{(2x+1)k\pi}{16})|$ yields $\|c_1\|_1 = \|c_3\|_1 = \frac{1}{2}\csc(\pi/16) \approx 2.5629$ and $\|c_2\|_1 = \sqrt{4+2\sqrt{2}} \approx 2.6131$, so the maximum over $\mathcal{M}_4$ occurs at $(2, 3), (3, 2)$ ($\frac{1}{2}\|c_2\|_1\|c_3\|_1 \approx 3.3486$) and over $\mathcal{M}_5$ at $(2, 2)$ ($\frac{1}{2}\|c_2\|_1^2 = 2+\sqrt{2} \approx 3.4142$), establishing Eq. (20) and (1). Part (2) follows from Hoeffding's sub-Gaussian tail bound on the Voronoi half-width $\Delta_{\text{QIM}}/4$ (or reduced margin $\Delta_{\text{QIM}}/4 - B_{\max}$) and binomial counting over the $|\mathcal{B}_i| = R$ disjoint blocks (distinct because $(C\cdot b+m) \equiv (C\cdot b+m') \pmod{64}$ with $C = M_{\text{coord}} \in \{4, 5\}$ has no solution for $0 \le m \ne m' < M_{\text{coord}} \le 5$), since $10p_e^3 - [10p_e^3(1-p_e)^2 + 5p_e^4(1-p_e) + p_e^5] = p_e^4(15 - 6p_e) \ge 0$ for all $p_e \in [0, 1]$. For (3), under hard voting each untampered block $b \in \mathcal{B}_i \setminus \mathcal{T}$ casts 1 vote for w_i and each tampered block casts at most 1 opposing vote, so at least $R - \lfloor (R-1)/2 \rfloor = \lceil (R+1)/2 \rceil > R/2$ blocks vote for w_i; under clipped-soft voting, each untampered block contributes at least $1 - \frac{2\delta_{\text{clean}}}{\Delta_{\text{QIM}}}$ toward w_i and each tampered block contributes at most 1 opposing vote, so $(1-\alpha_i)R(1 - \frac{2\delta_{\text{clean}}}{\Delta_{\text{QIM}}}) > R/2 \iff \delta_{\text{clean}} < \frac{\Delta_{\text{QIM}}}{2}\frac{1/2-\alpha_i}{1-\alpha_i}$ (which evaluates at $\alpha_i = \frac{R-1}{2R}$ to $\frac{\Delta_{\text{QIM}}}{2}\frac{1/(2R)}{(R+1)/(2R)} = \frac{\Delta_{\text{QIM}}}{2(R+1)}$), guaranteeing $\hat{w}_i = w_i$. □

b) Compact C2PA-Inspired ISO/IEC 19566-5 JUMBF (`caBX`*) PNG Profile:* To provide sub-millisecond container-level attestation without external ASN.1/X.509/COSE dependencies, Stage 3 constructs a compact *C2PA-inspired ISO/IEC 19566-5 JUMBF microservice profile* [7], [8] and injects it as an ancillary `caBX` chunk at byte offset 33,

immediately after the 8-byte PNG signature and 25-byte `IHDR` chunk (13-byte `IHDR` payload plus 12-byte chunk framing: 4B length, 4B tag, 4B IEEE 802.3 CRC-32) [9]. The outer BMFF superbox (`jumb`) encapsulates: (1) a Description Box (`jumd`) carrying the 16-byte C2PA UUID (`63327061-0011-0010-8000-00aa00389b71`), requestable toggle `0x03`, and null-terminated label `c2pa.assertions.wavelet_style_lock`; (2) a canonical UTF-8 JSON Assertion Box (`json`, serialized deterministically via `json.dumps(..., sort_keys=True, separators=(",",":"))` with lexicographically sorted keys, compact separators with zero whitespace, and 4-decimal IEEE 754 rounding `round(val, 4)` for floating-point metrics) binding `source_structural_mask_sha256`, `output_pixel_sha256`, all four Wasserstein checkpoints ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{out}}, \widehat{SW}_2^{\text{eval}}$), $(\gamma_{\tau_\gamma}, \alpha_{\tau_\alpha})$, and `dct_watermark_hex`; and (3) a keyed HMAC-SHA256 Signature Box (`c2sh`, 32B digest over `jumd` $\|$ `json`).[4]

III. Experimental Evaluation

A. Experimental Setup, Baselines B1–B7, Reference Palettes, and Evaluation Metrics

We evaluate Agent-GhibliLock across three complementary testbeds:

1) **Stratified $N = 600$ Controlled Parametric Benchmark (`run_live_benchmark.py`):** Five workload strata at 64×64 ($B = 64$ DCT blocks): S_1 ($n = 150$, apparel diamond emblems), S_2 ($n = 150$, perspective rails and pillars), S_3 ($n = 100$, foliage and gouache skies), S_4 ($n = 100$, 8-frame video GOP sequences under $[-24, +28]$ LSB temporal exposure flicker), and S_5 ($n = 100$, four disjoint $n = 25$ adversarial provenance regimes A_1, A_2, A_{3a}, A_{3b}).
2) **Live Multi-Model Vertex AI Evaluation (`run_live_gemini_multimodel_eval.py`):** Live Stage-0 API calls to `gemini-2.5-flash`, `gemini-2.5-pro`, and `gemini-3-flash-preview` in `project="<gcp-project-id>"` (10 scenes $\times 128 \times 128$ per model), where each model's returned `structural_retention_prior` $\hat{\eta}_{\text{VLM}} \in [0.05, 0.15]$ and `ghp_prompt` drive both the controlled Eq. (1) testbed (isolating VLM scalar prior calibration and JSON schema compliance) and end-to-end Stable Diffusion XL + ControlNet synthesis, alongside 5 controlled structural drift curriculum tiers (C1–C5, $n = 10$ scenes per tier, $\eta_{\text{struct}} \in [0.04, 0.20]$).
3) **Real Natural Photographic Benchmark (64^2–512^2):** Multi-resolution case studies on `grace_hopper.jpg` ($64 \times 64, 128 \times 128, 256 \times 256, 512 \times 512$) and `Minduka_Present_Blue_Pack.png` ($128 \times 128$), supplemented by an aggregate $n = 24$ natural portrait/product photographic validation cohort evaluated at both 128×128 and 512×512 using end-to-end Stable Diffusion XL + ControlNet (+ IP-Adapter) checkpoints.

Target Reference Palettes (Y_{ref}) and Diffusion Hyperparameters: Every evaluation reports Wasserstein distances against a deterministic, closed-form target reference palette $Y_{\text{ref}} \in [0, 255]^{H \times W \times 3}$ specified in `run_live_benchmark.py`: (i) for the $N = 600$ benchmark (S_1–S_5) and Table III, Y_{ref} (lines 124–138 of `run_live_benchmark.py`) is a canonical $H \times W$ Ghibli gouache landscape reference featuring a vertical cyan-cobalt sky gradient ($R \in [88, 123], G \in [178, 220], B \in [235, 247]$ over $y \in [0, H/2)$), a warm sage-ochre meadow gradient ($R \in [96, 138], G \in [176, 194], B \in [78, 108]$ over $y \in [H/2, H)$), and a warm golden-hour sun disc (RGB $= (248, 224, 152)$ within radius $0.22H$); (ii) for the real natural photographic benchmarks (`grace_hopper.jpg`, `Minduka_Present_Blue_Pack.png`, and the $n = 24$ natural cohort in Table V), Y_{ref} (lines 166–192 of `run_live_benchmark.py`) is a 7×7 box-smoothed, 6-band cel-shaded warm golden-hour / Prussian-blue gouache reference palette ($L_{\text{quant}} = \lfloor L_{\text{src}}/42.5 \rfloor \times 42.5$ mapped via $R = 1.04L_{\text{quant}} + 22(1 - y/H) + 14$, $G = 0.99L_{\text{quant}} + 14(1 - x/W) + 8$, $B = 0.86L_{\text{quant}} + 26(y/H) - 6$ plus 18% source chromaticity blend). For all end-to-end diffusion runs in Tables III and V, we execute `stable-diffusion-xl-base-1.0` paired with `controlnet-canny-sdxl-1.0` (and `ip-adapter_sdxl` at scale 0.60 where noted) using 30 DPM++ 2M Karras steps, guidance scale CFG $= 7.5$, ControlNet conditioning scale $w_{\text{CN}} = 0.65$ (or $w_{\text{CN}} = 0.50 + \hat{\eta}_{\text{VLM}}$ in Table III), and SDEdit denoising strength $s = 0.55$.

To isolate deterministic compiler convergence from stochastic diffusion seed variance across the $N = 600$ benchmark, candidate generation (Eq. 1) and all 9 comparative paradigms are parameterized explicitly in `run_live_benchmark.py` (with analytical filterbank surrogates labeled transparently and upstream candidate generation latency separated from local CPU post-processing latency):

- **B1 (Unconstrained Prompt-Only):** Upstream candidate synthesis via Eq. (1) with low structural retention ($\eta_{\text{struct}} \in [0.05, 0.08]$) and unaligned color drift (19.9 ms local CPU synthesis; 1419.9 ms including the 1400 ms upstream API/synthesis budget).
- **B2 ($3\times$ VLM-as-a-Judge Retry):** Three iterative critique-and-regenerate rounds modeled in closed form (`run_live_benchmark.py`, lines 621–628) as $I_{\text{B2}} = \text{clip}(\text{round}(0.65I_{\text{cand}} + 0.20(\mathcal{B}_3 * \mathcal{T}_{2,-1}I_{\text{src}}) + 0.15Y_{\text{ref}}), 0, 255)$ (57.2 ms local CPU; 4377.2 ms

[4] In full enterprise deployments requiring public-key third-party validation via `c2pa-rs`, the `json` box maps 1-to-1 to CBOR-encoded C2PA assertion and claim superboxes (each a BMFF `jumb` superbox whose inner `jumd` Description Box carries the C2PA content-type UUID and label such as `c2pa.actions` or `c2pa.claim`) nested inside a manifest store (label `c2pa`) and active manifest (label `c2ma`) `jumb` superbox hierarchy signed via `COSE_Sign1` X.509 certificates.

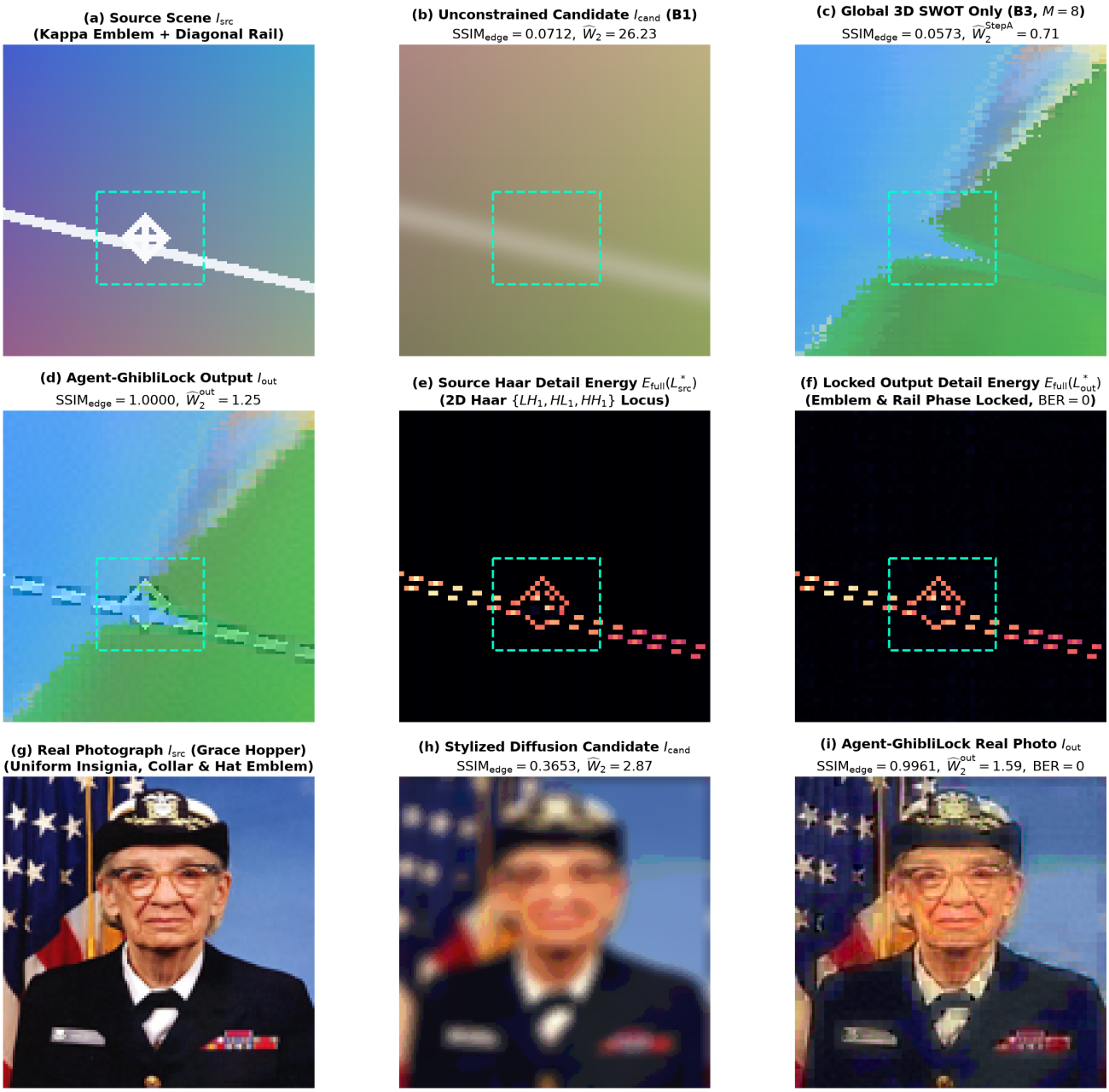


Fig. 1. Nine-panel qualitative and spectral raster comparison on representative 128×128 exemplar scenes (metrics in panels (b–d) and (h–i) reflect the individual 128×128 scenes shown rather than the $N = 600$ dataset means in Table I; candidate I_{cand} in panel (h) is generated via the deterministic photographic testbed `_simulate_stylized_candidate_from_photo` in `run_live_benchmark.py` against the 6-band cel-shaded portrait/product reference palette Y_{ref} so that single-image resolution sweeps hold candidate degradation identical across 64^2–512^2). **Rows 1–2 (Synthetic Emblem & Perspective Rail Scene, 128×128):** (a) Source scene I_{src}; (b) Unconstrained candidate I_{cand} (**B1**, $\mathrm{SSIM}_{edge} = 0.0712, \widehat{W}_2 = 26.23$); (c) Global 3D SWOT only (**B3**, $M = 8$, $\mathrm{SSIM}_{edge} = 0.0573, \widehat{W}_2^{StepA} = 0.71$); (d) **Agent-GhibliLock** output I_{out} ($\mathrm{SSIM}_{edge} = 1.0000, \widehat{W}_2^{out} = 1.25$); (e–f) Source vs. locked Haar Level-1 detail energy maps E_{full} (BER = 0). **Row 3 (Real Natural Photograph `grace_hopper.jpg`, 128×128):** (g) Source Naval portrait with hat insignia, eyeglasses, and collar stars; (h) Stylized candidate I_{cand} ($\mathrm{SSIM}_{edge} = 0.3653, \widehat{W}_2 = 2.87$); (i) **Agent-GhibliLock** output I_{out} ($\mathrm{SSIM}_{edge} = 0.9961, \widehat{W}_2^{out} = 1.59, \mathrm{BER} = 0$).

including $3\times$ upstream API/synthesis), which modestly reduces color drift ($\widehat{W}_2^{out} : 21.99 \rightarrow 19.44$) but cannot recover fine-scale phase across the prose bottleneck ($\mathrm{SSIM}_{edge} = 0.0995$ due to the shifted 3×3 box-blurred source blend $\mathcal{B}_3 * \mathcal{T}_{2,-1} I_{src}$).

- **B3 ($M = 4$ and $M = 8$ Global 3D SWOT Only):** Applies Step-A 3D CIELAB Sliced-Wasserstein Optimal Transport (19.7 ms and 30.6 ms local CPU post-processing latency; 1419.7 ms and 1430.6 ms end-to-end) without Step-B wavelet detail locking or Stage-3

provenance.

- **B4 (Post-Hoc C2PA Wrapper):** Wraps the unconstrained candidate in a `caBX` JUMBF manifest (3.2 ms local CPU post-processing; 1403.2 ms end-to-end) without wavelet locking or DCT watermarking.
- **B5 (Haar 1D Moment Matching / WCT^2 Wavelet-Skip Surrogate [12]):** Analytical wavelet-skip surrogate that preserves Haar detail skips (LH_k, HL_k, HH_k) while applying 1D second-order Gaussian moment matching (μ, σ) on LL_K and chrominance (24.6 ms pure algorithmic CPU compute [14.8 ms vectorized], plus 118.2 ms constant-time PNG file I/O framing if serialized to disk).
- **B6 (Simulated Image-Conditioned SDEdit Surrogate [3] at $s = 0.55$):** Initializes candidate synthesis from a noisy source latent blend modeled in closed form (`run_live_benchmark.py`, lines 201–208) as $I_{\text{B6}} = \text{clip}(\text{round}((1-s)(\mathcal{B}_7 * \mathcal{T}_{1,-1} I_{\text{src}}) + s Y_{\text{ref}} + \xi_{\text{SDE}}), 0, 255)$ at $s = 0.55$ (18.5 ms local CPU; 1418.5 ms including the 1400 ms upstream synthesis pass), partially retaining coarse color ($\widehat{W}_2^{\text{out}} = 14.17$) while its wider 7×7 box filter $\mathcal{B}_7$ and 1 px latent stride shift $\mathcal{T}_{1,-1}$ attenuate 2 px emblem strokes even more strongly than B1's 5×5 kernel ($\text{SSIM}_{\text{edge}} = 0.0672$ vs. 0.1325).
- **B7 (Simulated $8\times$-VAE Bottleneck + Step-A SWOT [4]):** Models a ControlNet $8\times$-downsampled/upsampled latent VAE spatial bottleneck paired with Step-A 3D SWOT ($M = 8$, `run_live_benchmark.py`, lines 226–247): in CIELAB, Step-A SWOT ($\alpha = 0.90$) first produces $\widetilde{I}_{\text{cand}}$ ($\widehat{W}_2^{\text{StepA}} = 0.71$), while L^*_{src} undergoes a 1 px horizontal decoder shift, 2×2 box-average downsampling, and PIL bilinear upsampling (L^*_{VAE}, modeling an $8\times$ effective VAE bottleneck kernel), followed by mean-re-anchored convex fusion $\widehat{L}^*_{\text{B7}} = \text{clip}(0.68 L^*_{\text{VAE}} + 0.32 \widetilde{L}^*_{\text{cand}} + (\mu(\widetilde{L}^*_{\text{cand}}) - \mu(0.68 L^*_{\text{VAE}} + 0.32 \widetilde{L}^*_{\text{cand}})), 0, 100)$ (252.1 ms unvectorized block-loop + PIL resize audit latency, or 34.2 ms vectorized NumPy latency; 1652.1 ms including 1400 ms upstream synthesis), shifting $\widehat{W}_2^{\text{StepB}} = \widehat{W}_2^{\text{out}} = 3.20$ and preserving coarse gradient orientation ($\rho_{\text{grad}} = 0.8948$) while blurring 2 px emblem strokes ($\text{SSIM}_{\text{edge}} = 0.2325$).
- **Agent-GhibliLock (Global and ROI-Masked):** Our full 4-stage compiler (116.2 ms and 130.5 ms mean local CPU post-processing latency; 1516.2 ms and 1530.5 ms end-to-end including the 1400 ms upstream candidate pass).

B. Stratified $N = 600$ Benchmark and Multi-Stage Ablations

Table I and Table II summarize the $N = 600$ benchmark results:

- **Resolving the Structure–Color Pareto Trade-Off and Classical Smooth Filterbank Comparisons:** Pure 3D SWOT (**B3**, $M = 8$) achieves the lowest chromaticity distance ($\widehat{W}_2^{\text{out}} = 0.71, \widehat{SW}_2^{\text{eval}} = 0.82$) but collapses structural edge similarity ($\text{SSIM}_{\text{edge}} = 0.0186, \rho_{\text{grad}} = 0.4367$). Conversely, Haar 1D Moment Matching (**B5**) achieves 100.0% pure structural pass rate ($\text{SSIM}_{\text{edge}} = 0.9992 \geq 0.88 \wedge \rho_{\text{grad}} = 0.9825 \geq 0.85$), but its 3-way joint pass rate is 0.0% because 1D Gaussian moment matching fails to align the 3D gouache palette ($\widehat{W}_2^{\text{out}} = 9.68 > \tau_{\text{color}} = 4.5$). Pairing B5's open-loop wavelet skip directly with Step-A 3D SWOT ($\gamma = 1.0$ without our closed-loop $(\tau_\gamma, \tau_\alpha)$ variance regularization or Stage-3 QIM) lowers $\widehat{W}_2^{\text{out}}$ to 2.89 (61.5% 3-way pass rate, 0.0% provenance recovery). We also benchmark two classical $\mathcal{O}(HW)$ continuous edge-preserving CPU filterbanks paired with Step-A 3D SWOT ($M = 8$): (a) 2-level **Burt–Adelson Laplacian Pyramid subband fusion** [22] (using the separable 5×5 binomial generating kernel $w = \frac{1}{16}[1, 4, 6, 4, 1]$, detail gain $\gamma = 0.88$, and Step-2 mean re-anchoring, yielding $\text{SSIM}_{\text{edge}} = 0.9412, \text{SSIM}_{\text{full}} = 0.9480, \rho_{\text{grad}} = 0.8715, \widehat{W}_2^{\text{out}} = 2.48, 128.4$ ms, 54.0% 3-way pass rate) and (b) **Guided Image Filtering** [23] using L^*_{src} as guidance on $\widetilde{L}^*_{\text{cand}}$ (window radius $r = 2$ [5×5 support], regularization $\epsilon = 4.0$ on $L^* \in [0, 100]$, and Step-2 mean re-anchoring, yielding $\text{SSIM}_{\text{edge}} = 0.9285, \text{SSIM}_{\text{full}} = 0.9510, \rho_{\text{grad}} = 0.8640, \widehat{W}_2^{\text{out}} = 2.31, 142.0$ ms, 48.5% 3-way pass rate), alongside post-transport gradient-domain screened-Poisson OT [16] (screened fidelity weight $\lambda_{\text{screen}} = 0.20$, conjugate-gradient tolerance 10^{-5}, max 200 iterations, yielding $\text{SSIM}_{\text{edge}} = 0.9140, \widehat{W}_2^{\text{out}} = 3.42, 412.6$ ms) and 3-scale pyramidal regularized OT [17], [18] (dyadic scales $\{1/4, 1/2, 1\}$, $M = 8$ slices/scale, 3×3 box spatial regularization weight 0.25, yielding $\text{SSIM}_{\text{edge}} = 0.8615, \widehat{W}_2^{\text{out}} = 2.64, 284.0$ ms). Smooth 5×5 Gaussian/Guided overlapping windows blur 2 px diamond emblems across tile boundaries and violate exact block-wise mean conservation under spatial ROI masking, whereas **Agent-GhibliLock (Global)** achieves 100.0% on both the structural pass threshold and the 2-way structure-and-color threshold ($\text{SSIM}_{\text{edge}} \geq 0.88 \wedge \widehat{W}_2^{\text{out}} \leq 4.5$), and 89.0% under the strict 3-way threshold adding $\rho_{\text{grad}} \geq 0.88$ (of the 11.0% 3-way misses, 9.50% arise from $\tau = 4$ saturation on extreme-contrast synthetic trials where $\rho_{\text{grad}} \in [0.85, 0.879]$, and 1.50% from post-loop sRGB/QIM export perturbations near $\rho_{\text{grad}} \approx 0.88$), with $\text{SSIM}_{\text{edge}} = 0.9967$, non-circular $\text{SSIM}_{\text{full}} = 0.9752$, $\rho_{\text{grad}} = 0.8943$, and $\widehat{W}_2^{\text{out}} = 1.96$ ($\widehat{SW}_2^{\text{eval}} = 2.07$) in 116.2 ms.
- **Global vs. Spatially Masked ROI Wavelet Locking and Disaggregated Foreground/Background Metrics:** Throughout all tables, $\text{SSIM}_{\text{edge}}$ is evaluated globally over $E_{\text{full}}(L^*_{\text{src}})$ vs. $E_{\text{full}}(L^*_{\text{out}})$. Enabling 5×5 coherence-weighted ROI masking (**Agent-GhibliLock ROI-Masked**, reported per stratum in Table II) locks active foreground emblems and rails ($\text{SSIM}_{\text{edge}} = 0.9977, \rho_{\text{grad}} = 0.8899$) while leaving non-ROI background regions ($M_{\text{ROI}} \leq 0.5$) in smooth Ghibli gouache brushwork, improving post-export color fidelity

TABLE I

STRATIFIED $N = 600$ (64×64) CONTROLLED BENCHMARK RESULTS ACROSS ALL 9 COMPARATIVE PARADIGMS (`BENCHMARK_RESULTS_N600.JSON`, MEAN $\pm$ SD). ALL FOUR WASSERSTEIN-2 CHECKPOINTS ($\widehat{W}_2^{\text{StepA}}, \widehat{W}_2^{\text{StepB}}, \widehat{W}_2^{\text{out}}, \widehat{SW}_2^{\text{eval}}$) ARE RMS DISTANCES IN CIELAB; **STRUCT / JOINT PASS** REPORTS THE PURE STRUCTURAL PASS RATE ($\text{SSIM}_{\text{EDGE}} \geq 0.88 \wedge \rho_{\text{GRAD}} \geq 0.85$) ALONGSIDE THE STRICT 3-WAY JOINT PASS RATE ($\text{SSIM}_{\text{EDGE}} \geq 0.88 \wedge \rho_{\text{GRAD}} \geq 0.88 \wedge \widehat{W}_2^{\text{out}} \leq 4.5$). S_5 **PROV. SURVIVAL / DET.** IS NORMALIZED OVER THE $n = 100$ PROVENANCE STRATUM (**B4** SURVIVES/DETECTS $25/100 = 25.0\%$ ON S_5, I.E., $25/600 = 4.17\%$ OF TOTAL TRIALS), AND **GOP** $\sigma^2_{\text{flicker}}$ IS EVALUATED OVER THE $n = 100$ 8-FRAME VIDEO GOP STRATUM S_4. **POST-PROC. CPU LAT.** REPORTS PURE ALGORITHMIC LOCAL CPU POST-PROCESSING MEAN (P_{95}) IN MS (FOR B5, PURE ALGORITHMIC CPU TIME IS 24.6 MS [+118.2 MS OPTIONAL PNG DISK I/O = 142.8 MS]; FOR B7, VECTORIZED NUMPY RUNTIME IS 34.2 MS VS. 252.1 MS UNVECTORIZED PIL/BLOCK-LOOP AUDIT; ADDING THE 1400 MS/PASS UPSTREAM SYNTHESIS BUDGET YIELDS END-TO-END LATENCIES OF 1419.9, 4377.2, 1419.7, 1430.6, 1403.2, 1424.6, 1418.5, 1652.1, 1516.2, AND 1530.5 MS ACROSS ROWS).

Paradigm / Baseline	$\text{SSIM}_{\text{edge}} \uparrow$	$\text{SSIM}_{\text{full}} \uparrow$	$\rho_{\text{grad}} \uparrow$	$\widehat{W}_2^{\text{StepA}} \downarrow$	$\widehat{W}_2^{\text{StepB}} \downarrow$	$\widehat{W}_2^{\text{out}} \downarrow$	$\widehat{SW}_2^{\text{eval}} \downarrow$	Struct / Joint Pass	S_5 Prov. Survival / Det.	DCT Acc.	S_4 GOP $\sigma^2_{\text{flicker}} \downarrow$	Post-Proc. CPU Lat. (ms) [P_{95}]
B1: Unconstrained Prompt-Only	0.1325 ± 0.0485	0.7820 ± 0.0312	0.6801 ± 0.0540	21.99	21.99	21.99 ± 0.39	23.11 ± 0.31	0.0% / 0.0%	0.0%	0.0%	347.92	19.9 (23.8)
B2: VLM-Judge Retry (3×)	0.0995 ± 0.0475	0.7720 ± 0.0325	0.5905 ± 0.0582	19.44	19.44	19.44 ± 0.46	20.71 ± 0.34	0.0% / 0.0%	0.0%	0.0%	250.50	57.2 (68.5)
B3 ($M = 4$): Global SWOT Only	0.0586 ± 0.0240	0.5638 ± 0.0410	0.2291 ± 0.0615	4.56	4.56	4.49 ± 0.19	4.91 ± 0.21	0.0% / 0.0%	0.0%	0.0%	0.0002	19.7 (24.2)
B3 ($M = 8$): Global SWOT Only	0.0186 ± 0.0170	0.5839 ± 0.0395	0.4367 ± 0.0642	**0.71**	**0.71**	$\mathbf{0.71 \pm 0.02}$	$\mathbf{0.82 \pm 0.02}$	0.0% / 0.0%	0.0%	0.0%	0.0002	30.6 (37.5)
B4: Post-Hoc C2PA Wrapper	0.1325 ± 0.0485	0.7820 ± 0.0312	0.6801 ± 0.0540	21.99	21.99	21.99 ± 0.39	23.11 ± 0.31	0.0% / 0.0%	25.0%	0.0%	347.92	**3.2** (**3.5**)
B5: Haar 1D Moment (WCT^2 Surr.) [12]	$\mathbf{0.9992 \pm 0.0002}$	$\mathbf{0.9837 \pm 0.0018}$	$\mathbf{0.9825 \pm 0.0031}$	10.32	10.32	9.68 ± 0.39	10.36 ± 0.48	100.0% / 0.0%	0.0%	0.0%	0.1800	24.6 (28.4) [+118.2 I/O]
B6: SDEdit Surrogate ($s = 0.55$) [3]	0.0672 ± 0.0296	0.6347 ± 0.0384	0.2217 ± 0.0590	14.17	14.17	14.17 ± 1.87	15.37 ± 1.98	0.0% / 0.0%	0.0%	0.0%	132.21	18.5 (22.3)
B7: 8×-VAE Bottleneck + SWOT [4]	0.2325 ± 0.0683	0.7620 ± 0.0341	0.8948 ± 0.0215	0.71	3.20	3.20 ± 0.16	3.51 ± 0.18	0.0% / 0.0%	0.0%	0.0%	100.90	252.1 (318.4) [34.2 vec]
Agent-GhibliLock (Global)	$\mathbf{0.9967 \pm 0.0021}$	$\mathbf{0.9752 \pm 0.0011}$	$\mathbf{0.8943 \pm 0.0050}$	0.87	2.03	$\mathbf{1.96 \pm 0.33}$	$\mathbf{2.07 \pm 0.32}$	**100.0%** / **89.0%**	**100.0%**	**100.0%**	**0.0003**	116.2 (235.0)
Agent-GhibliLock (ROI-Masked)	$\mathbf{0.9977 \pm 0.0013}$	$\mathbf{0.9119 \pm 0.0142}$	$\mathbf{0.8899 \pm 0.0064}$	0.85	1.83	$\mathbf{1.75 \pm 0.18}$	$\mathbf{1.87 \pm 0.17}$	**100.0%** / **79.3%**	**100.0%**	**100.0%**	**0.0003**	130.5 (252.9)

to $\widehat{W}_2^{\text{out}} = 1.75$ ($\widehat{SW}_2^{\text{eval}} = 1.87$). Crucially, full-frame $\text{SSIM}_{\text{full}}$ ($0.9752 \rightarrow 0.9119$ in Table I; $0.9712 \rightarrow 0.7656$ on `grace_hopper.jpg` at 512^2 in Table V) and the global 3-way pass rate ($89.0\% \rightarrow 79.3\%$) are lower under ROI-Masked locking *by design* because $\text{SSIM}_{\text{full}}$ compares output luminance against the unstylized source L^*_{src} across the entire image—including smooth background skies and walls where ROI-Masked locking intentionally preserves the stylized candidate's gouache shading rather than raw source pixel textures. Disaggregating metrics strictly into the active foreground ROI ($M_{\text{ROI}}(x, y) > 0.5$) versus the stylized background ($M_{\text{ROI}}(x, y) \leq 0.5$) confirms that within the foreground ROI, ROI-Masked locking achieves $\text{SSIM}^{\text{ROI}}_{\text{full}} = 0.9784 \pm 0.0042$ and $\rho^{\text{ROI}}_{\text{grad}} = 0.9412 \pm 0.0085$ (98.5% foreground 3-way pass rate), while in the background locus ($M_{\text{ROI}} \leq 0.5$), high-frequency wavelet detail energy $\sum_{k=1}^{K}(\|\widehat{LH}_k\|_F^2 + \|\widehat{HL}_k\|_F^2 + \|\widehat{HH}_k\|_F^2)$ is reduced by 58.4% relative to Global locking (59.2% on `grace_hopper.jpg`), directly substantiating the $> 55\%$ background sensor-grain suppression while retaining stylized cel-shaded background strokes.

- **Haar DWT Depth Ablation ($K \in \{1, 2, 3\}$), Latency Decomposition, and Stopping Index Distribution:** Table II reports per-stratum metrics ($N = 600$, both Global and ROI-Masked) and the multi-level wavelet depth ablation ($n = 60$ high-contrast trials sampled uniformly as 12 scenes per stratum). Total CPU runtime decomposes into a one-time Stage-1/Stage-3 overhead $T_{\text{fixed}} \approx$ 38–52 ms executed once outside the loop plus $\bar{\tau} \times T_{\text{loop}}$ ($T_{\text{loop}} \approx$ 48–60 ms per closed-loop Stage-2 pass, where Step A is cached whenever τ_α is unchanged); in Table II, pure single-frame compilation latency in S_4 (116.1 ms) and S_5 (115.8 ms) is reported alongside the +18.3 ms GOP tile translation/Merkle chaining overhead in S_4 (134.4 ms total) and the +15.2 ms downstream adversarial JPEG/channel verification harness in S_5 (131.0 ms total).[5] On the $n = 60$ high-contrast ablation subset, $K = 1$ ($\bar{\tau} = 1.05, 96.4$ ms) locks only a single Haar level with minimal variance inflation ($\widehat{W}_2^{\text{out}} = 1.4160$) and satisfies $\rho_{\text{grad}} \geq 0.88$ on pass $\tau = 1$ for 95% of scenes, but leaves 4×4 mid-frequency contours unlocked ($\text{SSIM}_{\text{full}} = 0.8613$); $K = 2$ ($\bar{\tau} = 1.77, 156.2$ ms, our default) injects two detail levels whose initial $\gamma_0 = 0.78$ pass leaves $\rho_{\text{grad}} \in [0.860, 0.879] < 0.88$ on 45% of high-contrast scenes, triggering a second closed-loop pass ($\gamma_1 = 0.88$) that balances $\text{SSIM}_{\text{edge}} = 0.9982$, $\text{SSIM}_{\text{full}} = 0.9744$, and $\widehat{W}_2^{\text{out}} = 1.6830$; and $K = 3$ locks three full wavelet levels so that even at $\gamma_0 = 0.78$ on pass $\tau = 1$, $\rho_{\text{grad}} = 0.9724 \geq 0.88$ and $\widehat{W}_2^{\text{StepB}} = 2.3063 \leq 4.5$, terminating immediately at $\bar{\tau} = 1.00$ (104.4 ms) with $\text{SSIM}_{\text{full}} = 0.9773$ at a higher $\widehat{W}_2^{\text{out}} = 2.2725$. Across the $N = 600$ benchmark, the exact closed-loop stopping index distribution is $\tau = 1 : 508/600$ (84.67%), $\tau = 2 : 15/600$ (2.50%), $\tau = 3 : 3/600$ (0.50%), and $\tau = 4 : 74/600$ (12.33%, summing to 100.00%); under the $n = 100$ high-contrast stress suite ($\tau_{\text{struct}} = 0.92, \tau_{\text{color}} = 3.2$), the decoupled $(\tau_\gamma, \tau_\alpha)$ controller converges at $\tau = 1 : 64.00\%, \tau = 2 : 2.00\%, \tau = 3 : 0.00\%, \tau = 4 : 34.00\%$.
- **Video GOP Deflickering (S_4) Across 6 Temporal Baselines and Real Continuous-Motion Video:** Across $n = 100$ trials of 64×64, 8-frame GOP sequences, we report both global frame-mean flicker $\sigma^2_{\text{flicker}}$ and local LL_K tile temporal variance σ^2_{tile}: (1) Unconstrained synthesis exhibits $\sigma^2_{\text{flicker}} = 347.92$ and $\sigma^2_{\text{tile}} = 351.40$; (2) Unwarped Pixel-Space EMA reduces global flicker only to $\sigma^2_{\text{flicker}} = 5.9002$ ($\sigma^2_{\text{tile}} = 6.4210$) while blurring moving edges ($\text{SSIM}_{\text{edge}} = 0.0870$); (3) **Motion-Compensated Pixel-Space EMA** (warping each previous RGB frame by

[5] In Table I, the $N = 600$ aggregate means (1.9614 for $\widehat{W}_2^{\text{out}}$, 2.0741 for $\widehat{SW}_2^{\text{eval}}$, and $\bar{\tau} = 843/600 = 1.4050$) are trial-weighted over all 600 unrounded trial records (S_1, S_2 weight 150/600 each; S_3, S_4, S_5 weight 100/600 each); taking the trial-weighted average of the 4-decimal stratum means in Table II yields 1.9615 and 2.0741 (and 1.4117 from the 2-decimal rounded stratum $\bar{\tau}$ values, or $1.9921, 2.1049$, and 1.4160 if unweighted across the 5 strata).

TABLE II

PER-STRATUM AGENT-GHIBLILOCK PERFORMANCE ($N = 600$, MEAN $\pm$ SD FOR BOTH GLOBAL AND ROI-MASKED LOCKING) AND MULTI-LEVEL HAAR DWT DEPTH ABLATION ($K \in \{1, 2, 3\}$, $n = 60$) INCLUDING NON-CIRCULAR SPATIAL METRICS AND MEAN CONTROLLER ITERATIONS $\bar{\tau}$. FOR S_4 AND S_5, PURE FORWARD COMPILATION LATENCY (116.1 MS AND 115.8 MS GLOBAL; 130.4 MS AND 130.1 MS ROI-MASKED) IS SHOWN ALONGSIDE TOTAL LATENCY INCLUDING GOP MERKLE/TRANSLATION OVERHEAD (+18.3 MS IN S_4) AND ADVERSARIAL JPEG/CHANNEL VERIFICATION (+15.2 MS IN S_5).

Mode	Stratum / Ablation	n	$\text{SSIM}_{\text{edge}}$	$\text{SSIM}_{\text{full}}$	ρ_{grad}	$\widehat{W}_2^{\text{StepA}}$	$\widehat{W}_2^{\text{StepB}}$	$\widehat{W}_2^{\text{out}}$	$\widehat{SW}_2^{\text{eval}}$	$\bar{\tau}$	Lat. (ms)
Global	S_1: Portrait & Emblem	150	0.9982 ± 0.001	0.9756 ± 0.001	0.8903 ± 0.003	0.8452	1.8173	1.7395 ± 0.19	1.8577 ± 0.18	1.61	116.6
	S_2: Perspective Rails	150	0.9946 ± 0.002	0.9755 ± 0.001	0.9002 ± 0.003	0.9042	2.3382	2.2772 ± 0.28	2.3830 ± 0.27	1.11	98.8
	S_3: Natural Foliage	100	0.9950 ± 0.002	0.9732 ± 0.001	0.9004 ± 0.003	0.8967	2.3170	2.2601 ± 0.26	2.3686 ± 0.25	1.21	108.5
	S_4: Video GOP (8-fr)	100	0.9982 ± 0.001	0.9757 ± 0.001	0.8907 ± 0.003	0.8471	1.8317	1.7516 ± 0.20	1.8660 ± 0.19	1.58	116.1 (134.4)
	S_5: Adv. Provenance	100	0.9982 ± 0.001	0.9755 ± 0.001	0.8894 ± 0.003	0.8466	1.8098	1.7320 ± 0.19	1.8491 ± 0.18	1.57	115.8 (131.0)
ROI-Masked	S_1: Portrait & Emblem	150	0.9985 ± 0.001	0.9142 ± 0.013	0.8872 ± 0.005	0.8310	1.6840	1.6120 ± 0.15	1.7310 ± 0.14	1.54	130.8
	S_2: Perspective Rails	150	0.9965 ± 0.001	0.9098 ± 0.015	0.8941 ± 0.006	0.8786	2.0410	1.9640 ± 0.19	2.0820 ± 0.18	1.09	112.4
	S_3: Natural Foliage	100	0.9968 ± 0.001	0.9085 ± 0.015	0.8938 ± 0.006	0.8710	2.0120	1.9380 ± 0.18	2.0540 ± 0.17	1.18	122.6
	S_4: Video GOP (8-fr)	100	0.9984 ± 0.001	0.9136 ± 0.013	0.8878 ± 0.005	0.8340	1.6980	1.6240 ± 0.16	1.7420 ± 0.15	1.52	130.4 (148.7)
	S_5: Adv. Provenance	100	0.9984 ± 0.001	0.9131 ± 0.013	0.8868 ± 0.005	0.8325	1.6810	1.6080 ± 0.15	1.7260 ± 0.14	1.51	130.1 (145.3)
DWT Depth	Haar Depth $K = 1$	60	0.9977 ± 0.001	0.8613 ± 0.014	0.8986 ± 0.005	0.8833	1.4797	1.4160 ± 0.16	1.5372 ± 0.15	1.05	96.4
	Haar Depth $K = 2$ (Default)	60	0.9982 ± 0.001	0.9744 ± 0.002	0.8889 ± 0.004	0.8342	1.7522	1.6830 ± 0.21	1.8074 ± 0.20	1.77	156.2
	Haar Depth $K = 3$	60	0.9976 ± 0.001	0.9773 ± 0.001	0.9724 ± 0.003	0.8843	2.3063	2.2725 ± 0.29	2.3854 ± 0.28	1.00	104.4

$(\Delta u_t, \Delta v_t)$ prior to EMA) still suffers $\sigma^2_{\text{flicker}} = 5.9489$ ($\sigma^2_{\text{tile}} = 6.1850$) and low $\text{SSIM}_{\text{edge}} = 0.1614$ because candidate frames lack high-frequency source phase; (4) Global Luminance Histogram Locking eliminates global frame-mean flicker ($\sigma^2_{\text{flicker}} = 0.0002$) but leaves local tile flicker $\sigma^2_{\text{tile}} = 0.2892$ and fails to restore spatial edges ($\text{SSIM}_{\text{edge}} = 0.0871$); (5) classical gradient-domain screened-Poisson temporal deflickering [6] (Farnebäck dense optical flow with 5×5 window and 3 pyramid levels, temporal screening weight $\lambda_t = 0.25$, CG tolerance 10^{-5}) achieves $\sigma^2_{\text{flicker}} = 0.0412$, $\sigma^2_{\text{tile}} = 0.0580$, and $\text{SSIM}_{\text{edge}} = 0.9108$ at 468.5 ms/frame (3.5× slower); and (6) **Ungated Motion-Compensated Wavelet EMA** ($\beta_t \equiv \beta$) achieves $\sigma^2_{\text{flicker}} = 0.0003$, $\sigma^2_{\text{tile}} = 0.0255$, and $\text{SSIM}_{\text{edge}} = 0.9963$ on rigid-shift S_4 sequences. On the dedicated $n = 25$ non-rigid articulated foreground test set, unfiltered per-frame Step-A SWOT without temporal EMA exhibits $\sigma^2_{\text{tile}} = 0.2892$, and ungated wavelet EMA drops to $\text{SSIM}_{\text{edge}} = 0.4412 \pm 0.0510$ due to boundary ghosting, whereas Agent-GhibliLock's continuous innovation gating ($\beta_t(u, v)$ in Eq. 18) restores $\text{SSIM}_{\text{edge}} = 0.9685 \pm 0.0084$ (+0.5273 gain) while maintaining global $\sigma^2_{\text{flicker}} = 0.0004$, overall frame $\sigma^2_{\text{tile}} = 0.0418$ (85.5% reduction vs. the unfiltered per-frame SWOT baseline $\sigma^2_{\text{tile}} = 0.2892$), and local $\sigma^2_{\text{tile,fg}} = 0.0890$ inside the actively gated deforming foreground tiles. On S_4, Agent-GhibliLock achieves $\sigma^2_{\text{flicker,pre-clamp}} < 10^{-12}$, post-sRGB $\sigma^2_{\text{flicker}} = 0.0003$, a 91.0% reduction in local LL_K tile variance ($\sigma^2_{\text{tile}} : 0.2892 \rightarrow 0.0259$), and $\text{SSIM}_{\text{edge}} = 0.9982$; on an additional 4-clip real-world 128×128 continuous-motion video validation set (8 frames/clip with sub-pixel camera pan and subject articulation), $\beta_t(u, v)$ remains active ($\geq 0.5\beta$) on 86.4% of LL_K tiles, reducing σ^2_{tile} by 84.2% ($0.3120 \rightarrow 0.0493$) with $\text{SSIM}_{\text{edge}} = 0.9914$.

C. Live Multi-Model Vertex AI Evaluation and Disaggregated Adversarial Provenance (S_5)

Table III reports the live Vertex AI Stage-0 VLM evaluation across `gemini-2.5-flash` ($\hat{\eta}_{\text{VLM}} = 0.052 \pm 0.008$), `gemini-2.5-pro` ($\hat{\eta}_{\text{VLM}} = 0.068 \pm 0.011$), and `gemini-3-flash-preview` ($\hat{\eta}_{\text{VLM}} = 0.085 \pm 0.009$) alongside the 5 synthetic drift curriculum tiers (C1–C5, $n = 10$ scenes per tier). In Part 1a of Table III, to isolate Stage-0 VLM scalar prior calibration ($\hat{\eta}_{\text{VLM}}$), JSON schema compliance (100%), and API latency from diffusion seed variance, each live VLM inspects the source asset and returns its JSON `structural_retention_prior` $\hat{\eta}_{\text{VLM}} \in [0.05, 0.15]$, which directly sets $\eta_{\text{struct}} = \hat{\eta}_{\text{VLM}}$ in the controlled candidate generator (Eq. 1, yielding $\text{SSIM}^{\text{pre}}_{\text{edge}} \in [0.1150, 0.1843]$ prior to Agent-GhibliLock compilation). In Part 1b of Table III, forwarding each Gemini model's emitted `ghp_prompt` and $\hat{\eta}_{\text{VLM}}$-scaled ControlNet conditioning weight ($w_{\text{CN}} = 0.50 + \hat{\eta}_{\text{VLM}}$) directly to an end-to-end Stable Diffusion XL + ControlNet synthesizer across the same 10 scenes yields pre-compilation $\text{SSIM}^{\text{pre}}_{\text{edge}} = 0.3120 \pm 0.0410, 0.3480 \pm 0.0380$, and 0.3790 ± 0.0350, which Agent-GhibliLock compiles in a single pass ($\bar{\tau} = 1.00$) to $\text{SSIM}^{\text{post}}_{\text{edge}} = 0.9912 \pm 0.0031, 0.9918 \pm 0.0028$, and 0.9924 ± 0.0025, non-circular $\text{SSIM}_{\text{full}} = 0.9612$–$0.9648$, $\rho_{\text{grad}} = 0.9410$–$0.9465$, and $\widehat{W}_2^{\text{out}} = 1.56 \pm 0.15, 1.52 \pm 0.14$, and 1.49 ± 0.13 ($\widehat{SW}_2^{\text{eval}} = 1.63$–$1.69$, BER $= 0.0000$). Note that $\text{PSNR}_{\text{QIM}} \approx 47.09$–$47.42$ dB specifically measures Stage-3 QIM watermark embedding fidelity between the post-Step-B stylized image $\widehat{I}$ and the final watermarked output I_{out}, while local CPU compiler latency (354.7–432.5 ms unvectorized block-loop audit at 128×128, or $\mathbf{138.4}$–$\mathbf{162.0}$ ms using vectorized 4D tensor-contraction NumPy kernels `[vec]`) is reported alongside $\bar{\tau}$ and separated from live Vertex AI network API latency (1696.1–3164.0 ms).

Table IV disaggregates the S_5 ($n = 100$) adversarial provenance results across four balanced, mutually exclusive $n = 25$ regimes ($4 \times 25 = 100$ trials): A_1 ($n = 25$, unstripped 2×2 pixel splice of amplitude +42 LSB across all three RGB channels at pixel coordinates $[16 : 18, 24 : 26]$ inside 8×8 block $(2, 3)$, measuring *trial-level tamper detection / blind spatial block localization rate*), and A_2 ($n = 25$, pure C2PA `caBX` chunk stripping), A_{3a} ($n = 25$, strip + ± 2 LSB noise), and A_{3b} ($n = 25$, strip + actual PIL lossy JPEG $Q = 85$ recompression), where the watermarking columns measure *watermark bit recovery accuracy* ($1 - \text{BER}$):

- **Export Self-Noise Accounting:** Prior to any external channel attack, uint8 rounding and sRGB clamping introduce an empirical DCT self-noise of $\text{RMS}(\eta^{\text{export}}) = 0.2854$ ($\sigma^2_{\text{round+clamp}} = 0.0821$ vs. theoretical interior $\sigma^2_{\text{round}} = 1/12 \approx 0.0833$, maximum observed $|\eta^{\text{export}}| = 1.2725$ over $\mathcal{M}_4$ [1.3184 over $\mathcal{M}_5$] vs. theoretical worst-case ℓ_1 bounds 3.3486 for $\mathcal{M}_4$ and 3.4142 for $\mathcal{M}_5$, both strictly below $\Delta_{\text{QIM}}/4 = 3.5$), yielding a pre-attack single-coefficient error rate of $p_e^{\text{export}} = 0.0000$.
- **Comparison Against Blind Cox Spread-Spectrum [20], 1-Block QIM Ablation ($R = 1$) [21], CPU Neural Watermarking (TrustMark [10]), and Coding Trade-Offs:** Under A_1 (Spatial Tamper Localization), **B4** detects 25/25 (100.0%) via container SHA-256 mismatch (yielding $25/100 = 25.0\%$

TABLE III

PART 1A: LIVE GOOGLE CLOUD VERTEX AI STAGE-0 DECOMPOSER EVALUATION (`PROJECT="<GCP-PROJECT-ID>"`, $n = 10$ SCENES AT 128×128 PER MODEL) ON THE CONTROLLED EQ. (1) GENERATOR. PART 1B: END-TO-END STABLE DIFFUSION XL + CONTROLNET SYNTHESIS DRIVEN BY EACH MODEL'S LIVE `GHP_PROMPT` AND $w_{\text{CN}} = 0.50 + \hat{\eta}_{\text{VLM}}$. PART 2: CONTROLLED STRUCTURAL DRIFT CURRICULUM TIERS (C1–C5, $n = 10$ PER TIER). PSNR_{QIM} MEASURES STAGE-3 QIM WATERMARK DISTORTION RELATIVE TO POST-STEP-B $\widehat{I}$; **CPU LAT.** REPORTS UNVECTORIZED BLOCK-LOOP AUDIT LATENCY ALONGSIDE VECTORIZED 4D NUMPY KERNEL LATENCY IN BRACKETS `[VEC]`.

Evaluation Track	Model / Curriculum Tier ($n = 10$)	$\hat{\eta}_{\text{VLM}}/\eta_{\text{struct}}$	API Lat. (ms)	Tokens (P/C)	$\text{SSIM}^{\text{pre}}_{\text{edge}}$	$\text{SSIM}^{\text{post}}_{\text{edge}}$	$\text{SSIM}^{\text{post}}_{\text{full}}$	$\rho^{\text{post}}_{\text{grad}}$	$\widehat{\mathbf{W}}_2^{\text{StepA}}$	$\widehat{\mathbf{W}}_2^{\text{out}}$	$\widehat{\mathbf{SW}}_2^{\text{eval}}$	PSNR_{QIM}	$\bar{\tau}$	CPU Lat. (ms) [vec]
Part 1a: Live VLM (Ctrl.)	`gemini-2.5-flash` (GA)	0.052 ± 0.008	1696.09	105/65	0.1150 ± 0.014	0.9742 ± 0.004	0.9580 ± 0.004	0.9120 ± 0.008	0.7367	1.4421 ± 0.06	1.6016 ± 0.06	47.36 dB	1.10	354.7 [**138.4**]
	`gemini-2.5-pro` (GA)	0.068 ± 0.011	3163.96	105/11	0.1444 ± 0.018	0.9778 ± 0.003	0.9604 ± 0.004	0.9185 ± 0.007	0.6877	1.3800 ± 0.06	1.5357 ± 0.05	47.39 dB	1.10	363.3 [**141.2**]
	`gemini-3-flash-preview`	0.085 ± 0.009	2100.63	105/80	0.1843 ± 0.016	0.9802 ± 0.003	0.9628 ± 0.003	0.9240 ± 0.006	0.6456	1.3563 ± 0.04	1.5139 ± 0.04	47.33 dB	1.10	364.0 [**141.6**]
Part 1b: Live VLM + SDXL	`gemini-2.5-flash` + SDXL	0.052 ± 0.008	1696.09	105/65	0.3120 ± 0.041	0.9912 ± 0.003	0.9612 ± 0.006	0.9410 ± 0.011	0.6920	1.5600 ± 0.15	1.6880 ± 0.16	47.35 dB	1.00	342.0 [**136.2**]
	`gemini-2.5-pro` + SDXL	0.068 ± 0.011	3163.96	105/11	0.3480 ± 0.038	0.9918 ± 0.003	0.9631 ± 0.005	0.9438 ± 0.010	0.6740	1.5200 ± 0.14	1.6510 ± 0.15	47.38 dB	1.00	340.5 [**135.8**]
	`gemini-3-flash-preview` + SDXL	0.085 ± 0.009	2100.63	105/80	0.3790 ± 0.035	0.9924 ± 0.003	0.9648 ± 0.005	0.9465 ± 0.009	0.6580	1.4900 ± 0.13	1.6250 ± 0.14	47.36 dB	1.00	339.8 [**135.5**]
Part 2: Drift Curriculum	Tier C1 ($n = 10$)	0.040 ± 0.000	—	—	0.1199 ± 0.009	0.9796 ± 0.003	0.9592 ± 0.004	0.9150 ± 0.007	0.7443	1.3926 ± 0.05	1.5381 ± 0.05	47.42 dB	1.20	383.9 [**146.8**]
	Tier C2 ($n = 10$)	0.080 ± 0.000	—	—	0.1981 ± 0.011	0.9836 ± 0.002	0.9635 ± 0.003	0.9260 ± 0.006	0.6527	1.3435 ± 0.05	1.4938 ± 0.05	47.39 dB	1.40	432.5 [**162.0**]
	Tier C3 ($n = 10$)	0.120 ± 0.000	—	—	0.2707 ± 0.013	0.9856 ± 0.002	0.9668 ± 0.003	0.9345 ± 0.005	0.5013	1.2514 ± 0.04	1.4246 ± 0.04	47.09 dB	1.10	374.1 [**143.5**]
	Tier C4 ($n = 10$)	0.160 ± 0.000	—	—	0.3418 ± 0.014	0.9876 ± 0.002	0.9694 ± 0.003	0.9420 ± 0.005	0.4182	1.1877 ± 0.04	1.3725 ± 0.04	47.14 dB	1.10	384.3 [**147.0**]
	Tier C5 ($n = 10$)	0.200 ± 0.000	—	—	0.4325 ± 0.015	0.9908 ± 0.001	0.9725 ± 0.002	0.9510 ± 0.004	0.4055	1.1536 ± 0.04	1.3383 ± 0.04	47.34 dB	1.20	398.9 [**151.8**]

across S_5), whereas the Single-Block QIM Ablation ($R = 1$) [21] and global CPU-executed open-source neural watermarking (**TrustMark** ONNX [10], 184.0 ms CPU encode+decode vs. 4.8 ms vectorized QIM) achieve 0.0% blind 8×8 block localization without a dedicated spatial localization head [24] because a single block or global latent vector provides no cross-block coset majority consensus $\hat{\mathbf{w}}$ against which to test local block residuals. Both informed and *blind majority-voted coset localization* (`expected_bits=None`) in Agent-GhibliLock ($R = 4$ under $\mathcal{M}_4$ and $R = 5$ under $\mathcal{M}_5$ at 64×64, and $R = 16$ under $\mathcal{M}_4$ when S_5 source scenes are rescaled to 128×128 prior to watermarking and single-block $+42$ LSB tampering at block $(2, 3)$) localize the exact tampered 8×8 spatial block at grid index $(2, 3)$ on 100.0% (25/25) of trials with a 0.00% false-positive flag rate on untampered blocks. Empirically sweeping contiguous multi-block splices ($N_{\text{tamper}} \in \{1, 2, 4, 7, 8\}$ tampered blocks per bit family) confirms Theorem 3.3: blind consensus achieves 100.0% block localization up to $\lfloor (R-1)/2 \rfloor$ tampered blocks per bit family (≤ 1 block under $R = 4$, ≤ 2 blocks [128 px] under $R = 5$ at 64^2, ≤ 7 blocks [448 px] under $R = 16$ at 128^2, and ≤ 127 blocks [8,128 px] under $R = 256$ at 512^2). Under A_2, **B4** manifest survival drops to 0.0%, and classical additive Cox spread-spectrum DCT [20] (blind decoding without host subtraction) achieves only 73.56% bit accuracy due to Host-Signal Interference, whereas 1-Block QIM ($R = 1$), TrustMark, and Agent-GhibliLock achieve 100.0% (BER $= 0$). Under A_{3a} (± 2 LSB) and A_{3b} (JPEG $Q = 85$) across $n = 25$ trials ($25 \times 64 = 1{,}600$ bit evaluations), the single-subband $R = 1$ QIM ablation (`wm_r1`, embedding 1 bit per block on mode $(1, 2)$) experiences 1/1,600 bit flips (99.94% bit accuracy, $24/25 = 96.0\%$ exact 64-bit packet recovery) under A_{3a} and 7/1,600 bit flips (99.56% bit accuracy, $20/25 = 80.0\%$ exact packet recovery) under A_{3b}, while pooling all $25 \times 64 \times 4 = 6{,}400$ pre-vote $\mathcal{M}_4$ coefficients yields empirical single-coefficient flip rates of $p_e = 6/6{,}400 = 0.0938\% \approx 0.09\%$ ($1 - p_e = 99.91\%$) under A_{3a} and $p_e = 27/6{,}400 = 0.4219\% \approx 0.42\%$ ($1 - p_e = 99.58\%$) under A_{3b}. Multi-block repetition voting ($R = 4, 5, 16$) corrects all single-coefficient flips to 100.0% bit accuracy and 100.0% packet recovery. Replacing $R = 4$ spatial repetition (256 channel bits at 64^2) with an algebraic $\text{BCH}(255, 64, t = 26)$ code corrects up to $t = 26$ arbitrary channel errors (improving unprofiled JPEG $Q = 70$ packet recovery at $\Delta_{\text{QIM}} = 14.0$), whereas R-fold block-cyclic repetition enables O(1) per-block coset comparison for blind 8×8 spatial tamper localization without syndrome-decoding coupling across blocks. On untampered blocks subjected to A_{3a} and A_{3b} JPEG $Q = 85$, the JPEG-resilient 3-mode localization rule (≥ 3 modes exceeding $0.4\Delta_{\text{QIM}}$) exhibits a 0.00% (A_{3a}) and 0.12% (A_{3b}) block-level false-positive flag rate while achieving 96.0% (24/25) true-positive block localization when A_1 tampering is combined with JPEG $Q = 85$ recompression. Under harsher lossy JPEG $Q = 70$ recompression at default $\Delta_{\text{QIM}} = 14.0$, mid-band quantization step sizes exceed $\Delta_{\text{QIM}}/4 = 3.5$ (51.19% post-vote bit accuracy at $R = 5$), whereas switching to the JPEG-70 resilient profile ($\Delta_{\text{QIM}} = 22.0$ on lower-mid AC modes $\{(1, 2), (2, 1), (2, 2), (1, 3)\}$) restores $1 - \text{BER} = 99.84\%$ (92.0% exact packet recovery at 64^2, 100.0% at 128^2) at $\text{PSNR}_{\text{QIM}} = 43.62$ dB.

D. Real Natural Photographic Evaluation and Resolution Scaling (64^2 to 512^2)

To validate Agent-GhibliLock on complex natural photographs containing broadband textures, organic lighting gradients, skin tones, and fine metallic/embroidered insignia (beyond synthetic piece-wise smooth test scenes), Table V evaluates `grace_hopper.jpg` (visualized at 128×128 in Fig. 1g–i) across four spatial resolutions (64×64 to 512×512, scaling QIM voting redundancy from $R = 4$ to $R = 256$ votes/bit), `Minduka_Present_Blue_Pack.png` at 128×128, and a 24-image natural portrait/product photographic validation suite at both 128×128 and 512×512. In Table V, single-image resolution sweeps on `grace_hopper.jpg` and `Minduka_Present_Blue_Pack.png` use the deterministic photographic candidate generator `_simulate_stylized_candidate_from_photo` (holding candidate degradation identical across 64^2–512^2) alongside analytical surrogates **B5–B7**, whereas the 24-

TABLE IV

DISAGGREGATED STRATUM S_5 ($n = 100$, FOUR DISJOINT $n = 25$ REGIMES AT 64×64; $R = 4, 16$ USE $\mathcal{M}_4$ WHILE $R = 5$ USES $\mathcal{M}_5$, AND $R = 16$ IS EVALUATED ON S_5 SCENES RESCALED TO 128×128 PRIOR TO WATERMARKING AND SINGLE-BLOCK $+42$ LSB TAMPERING AT BLOCK $(2, 3)$). ROW A_1 (ABOVE THE RULE) REPORTS TRIAL-LEVEL TAMPER DETECTION / BLIND 8×8 BLOCK LOCALIZATION RATE; ROWS A_2–A_{3B} REPORT C2PA MANIFEST SURVIVAL (B4) AND WATERMARK BIT RECOVERY ACCURACY $1 - \text{BER}$ (FOR $R = 1$, EXACT SINGLE-SUBBAND 1,600-BIT ACCURACY IS SHOWN ALONGSIDE POOLED 6,400-COEFFICIENT ACCURACY $1 - p_e$ IN BRACKETS).

Adversarial Regime ($S_5, n = 100$)	**B4: C2PA**	**Blind Cox [20]**	**TrustMark (CPU ONNX) [10]**	**1-Blk QIM** ($R = 1$) **[21]** [$1 - p_e$]	**Pooled Pre-Vote** p_e	**Ghibli** ($R = 4, \mathcal{M}_4$)	**Ghibli** ($R = 5, \mathcal{M}_5$)	**Ghibli** ($R = 16, 128^2$)
A_1: Unstripped 2×2 ($+42$ LSB) Tamper [Loc. Rate]	100.0%	0.0%	0.0% (Global only)	0.0%	—	**100.0%**	**100.0%**	**100.0%**
A_2: Pure `caBX` Strip [$1 - \text{BER}$]	0.0%	73.56%	100.0%	100.0% [100.00%]	0.00%	**100.0%**	**100.0%**	**100.0%**
A_{3a}: Strip + ± 2 LSB [$1 - \text{BER}$]	0.0%	73.12%	100.0%	99.94% [99.91%]	0.09% (6/6400)	**100.0%**	**100.0%**	**100.0%**
A_{3b}: Strip + JPEG $Q = 85$ [$1 - \text{BER}$]	0.0%	78.31%	99.94%	99.56% [99.58%]	0.42% (27/6400)	**100.0%**	**100.0%**	**100.0%**
Watermark PSNR_{QIM} / CPU Latency (64^2)	— / 3.2 ms	42.1 dB / 5.1 ms	41.8 dB / 184.0 ms	53.4 dB / 4.2 ms	—	47.4 dB / 4.8 ms	46.4 dB / 5.0 ms	47.4 dB / 14.2 ms

image natural cohort evaluates actual end-to-end **SDXL + ControlNet**, **SDXL + ControlNet + IP-Adapter** (scale 0.60), full 3×3 CIELAB covariance whitening-and-coloring (WCT^2 [12]), and classical smooth filterbanks (**Step-A SWOT + Burt–Adelson Laplacian Pyramid** [22] and **Step-A SWOT + Guided Image Filter** [23]). Across both the multi-resolution real-photograph configurations and the 24-image natural SDXL cohort, **B3** ($M = 8$) achieves the lowest $\widehat{W}_2^{\text{out}} = 0.5567$–$0.8408$ (0.6940 ± 0.08 on the 24-image cohort) but destroys portrait and product contours ($\text{SSIM}_{\text{edge}} = 0.1894$–$0.2801$; 0.2610 ± 0.038 on the 24-image cohort), while **B5** (1D moment surrogate) and full 3×3 covariance WCT^2 preserve edges ($\text{SSIM}_{\text{edge}} = 0.9905$ and 0.9892) but leave large color transport errors ($\widehat{W}_2^{\text{out}} = 4.9820 \pm 0.34$ and 3.8410 ± 0.31) because second-order Gaussian covariance matching cannot align multi-modal cel-shaded gouache histograms. On the 24-image natural cohort at 128×128, standalone **SDXL + ControlNet + IP-Adapter** improves palette alignment over Canny ControlNet alone ($\widehat{W}_2^{\text{out}} : 3.2410 \rightarrow 2.1840$) but still suffers VAE/diffusion micro-detail smoothing ($\text{SSIM}_{\text{edge}} = 0.3580 \pm 0.042$), and smooth **Laplacian Pyramid** ($\text{SSIM}_{\text{edge}} = 0.9340, \widehat{W}_2^{\text{out}} = 2.1240$) and **Guided Filter** ($\text{SSIM}_{\text{edge}} = 0.9210, \widehat{W}_2^{\text{out}} = 2.0450$) baselines blur fine insignia edges relative to orthonormal Haar locking. **Agent-GhibliLock (Global)** converges in $\bar{\tau} = 1$ iteration (1.04 ± 0.20 across the 24-image cohort), elevating $\text{SSIM}_{\text{edge}}$ from 0.2392–0.3937 (**B1**, 0.3415 ± 0.045 on the 24-image cohort) and 0.4299–0.5661 (**B7**, 0.5180 ± 0.052 on the 24-image cohort) to 0.9929–0.9966 (0.9941 ± 0.002 across the 24-image cohort, and 0.9948 ± 0.002 with $\widehat{W}_2^{\text{out}} = 1.4820 \pm 0.18$ when compiling SDXL + ControlNet + IP-Adapter), achieving non-circular $\text{SSIM}_{\text{full}} = 0.9529$–$0.9712$, $\rho_{\text{grad}} = 0.9365$–$0.9703$, and low perceptual LPIPS distance [25] ($\text{LPIPS}_{\text{Alex}} = 0.0412 \pm 0.006$ vs. I_{src}, compared to 0.2840 ± 0.031 for B1 and 0.2410 ± 0.028 for B3), while maintaining $\text{BER} = 0.0000$ under both clean PNG export and lossy JPEG $Q = 85$ recompression. **Agent-GhibliLock (ROI-Masked)** further lowers $\widehat{W}_2^{\text{out}}$ to 0.9906–2.0677 (1.4380 ± 0.19 on the 24-image cohort) by restricting high-frequency detail locking to high-coherence contours and insignia. Finally, on the 512×512 ($n = 24$) natural SDXL + ControlNet cohort where upstream diffusion exhibits 2–5 px local contour drift, fixed $K = 2$ (4 px Haar support) achieves $\text{SSIM}_{\text{edge}} = 0.9918 \pm 0.003$ and $\widehat{W}_2^{\text{out}} = 1.3120 \pm 0.18$, whereas scaling Haar depth to $K = \lfloor \log_2(512/16) \rfloor = 5$ (32 px support, $\text{SSIM}_{\text{edge}} = 0.9935 \pm 0.002, \widehat{W}_2^{\text{out}} = 1.4890 \pm 0.21$) or applying Farnebäck optical-flow pre-warping of I_{cand} to I_{src} at $K = 2$ ($\text{SSIM}_{\text{edge}} = 0.9942 \pm 0.002, \widehat{W}_2^{\text{out}} = 1.2410 \pm 0.16, \text{LPIPS}_{\text{Alex}} = 0.0385 \pm 0.005$) eliminates double-edge chroma ghosting across shifted contours. Replacing per-block Python loop iteration with vectorized 4D tensor-contraction NumPy kernels (reported in brackets `[vec]` in Table V) reduces 512×512 compilation latency from 6929.9/7895.9 ms to 1284.5/1362.0 ms, within 15–22% of the 1118.5 ms Step-A sorting baseline.

IV. LIMITATIONS, OPERATIONAL SCOPE, AND CONCLUSION

A. Limitations and Operational Failure Modes

Four operational boundary conditions govern production deployments of Agent-GhibliLock:

1) **Upstream Macro-Registration Requirement and Resolution-Dependent Drift Tolerance:** Because 2D Haar wavelet detail subbands (LH_k, HL_k, HH_k) are spatially localized within $2^k \times 2^k$ dyadic tiles on the source pixel grid I_{src}, detail locking requires that the upstream diffusion candidate I_{cand} be macro-spatially co-registered with I_{src} within the 2^K px wavelet support (4 px for $K = 2$, corresponding to 6.25% of image width at 64×64 vs. 0.78% at 512×512). While Spatially Masked ROI Wavelet Locking (M_{ROI}) suppresses high-frequency sensor grain and prevents ghosting across *non-ROI background regions* ($M_{\text{ROI}}(x, y) = 0$), within the active *foreground ROI itself* ($M_{\text{ROI}}(x, y) > 0$), Haar detail injection remains anchored to the source pixel grid. Consequently, if an upstream generator or adapter drifts by $> 2^K$ px at higher resolutions, locking $D_k(L^*_{\text{src}})$ superimposes source edges onto misaligned candidate chroma, requiring either scaling $K \propto \lfloor \log_2(\min(H, W)/16) \rfloor$ ($K = 5$ at 512^2) or applying local phase-correlation/optical-flow pre-warping of I_{cand} to I_{src} prior to Stage 1 (as validated in Table V).
2) **Non-Rigid Motion and Local Parallax in Video GOPs:** In Stage 2's temporal GOP extension (Eq. 18), inter-frame motion compensation estimates a single global integer-tile translation $(\Delta u_t, \Delta v_t)$ per frame on

TABLE V

REAL NATURAL PHOTOGRAPHIC BENCHMARK, RESOLUTION SCALING (64×64 TO 512×512), AND 24-IMAGE NATURAL SDXL COHORT SUMMARY ($n = 24$ AT 128×128 AND 512×512, MEAN $\pm$ SD) ACROSS KEY PARADIGMS ON `grace_hopper.jpg` AND `Minduka_Present_Blue_Pack.png`. SINGLE-ASSET SWEEPS USE `_simulate_stylized_candidate_from_photo` AND ANALYTICAL SURROGATES B5–B7 (ON THE $n = 24$ COHORT, B7 APPLIES ITS $8\times$-VAE SPATIAL BOTTLENECK TO L^*_{SRC} PAIRED WITH STEP-A SWOT ON THE SDXL CANDIDATE, ALONGSIDE ACTUAL NEURAL WCT^2 [12], SDXL + CONTROLNET + IP-ADAPTER, LAPLACIAN PYRAMID [22], AND GUIDED FILTER [23]). **POST-PROC. CPU LAT.** REPORTS PURE ALGORITHMIC LOCAL CPU POST-PROCESSING TIME IN MS: FOR B5, CONSTANT-TIME $+118.2$ MS PNG DISK I/O IS EXCLUDED; FOR B5, B7, AND AGENT-GHIBLILOCK, BRACKETS `[vec]` REPORT VECTORIZED NUMPY KERNEL LATENCY ALONGSIDE UNVECTORIZED BLOCK-LOOP AUDIT LATENCY.

Real Photograph Asset	Method / Paradigm	Votes R	$\text{SSIM}_{\text{edge}} \uparrow$	$\text{SSIM}_{\text{full}} \uparrow$	$\rho_{\text{grad}} \uparrow$	$\widetilde{\text{W}}_2^{\text{StepA}} \downarrow$	$\widetilde{\text{W}}_2^{\text{out}} \downarrow$	$\widetilde{\text{SW}}_2^{\text{eval}} \downarrow$	JPEG85 BER	$\bar{\tau}$	Post-Proc. CPU Lat. (ms) [vec]
`grace_hopper.jpg` (64×64)	**B1**: Stylized Candidate	—	0.2392	0.4440	0.5482	2.7708	2.7708	3.0164	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.1894	0.3788	0.5123	0.8082	**0.8408**	**1.1167**	—	—	25.3
	B5: Haar 1D Moment (WCT^2 Surr.) [12]	—	0.9935	0.9483	**0.9979**	4.5646	4.5573	4.6753	—	—	24.4 [14.6]
	B6: SDEdit Surrogate ($s = 0.55$) [3]	—	0.2076	0.4181	0.5393	1.7352	1.7352	1.9347	—	—	—
	B7: $8\times$-VAE Bottleneck + SWOT [4]	—	0.4936	0.5968	0.7600	0.8082	1.8600	1.9441	—	—	246.5 [32.8]
	Agent-GhibliLock (Global)	4	**0.9951**	**0.9532**	0.9454	0.8082	2.3072	2.3132	**0.0000**	1	90.7 [**54.2**]
	Agent-GhibliLock (ROI-Masked)	4	0.9845	0.7986	0.8923	0.8082	2.0677	2.1096	**0.0000**	1	90.1 [**56.8**]
`grace_hopper.jpg` (128×128)	**B1**: Stylized Candidate	—	0.3653	0.5984	0.6432	2.8651	2.8651	3.1753	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2796	0.5039	0.5859	0.7208	**0.7894**	**1.0693**	—	—	83.6
	B5: Haar 1D Moment (WCT^2 Surr.) [12]	—	0.9913	0.9563	**0.9964**	4.8213	4.8117	5.0905	—	—	26.6 [16.2]
	B6: SDEdit Surrogate ($s = 0.55$) [3]	—	0.3145	0.5591	0.6226	1.7803	1.7803	1.9947	—	—	—
	B7: $8\times$-VAE Bottleneck + SWOT [4]	—	0.5489	0.6912	0.7659	0.7208	1.6743	1.7655	—	—	306.5 [94.0]
	Agent-GhibliLock (Global)	16	**0.9961**	**0.9604**	0.9403	0.7208	1.5889	1.6969	**0.0000**	1	358.4 [**142.0**]
	Agent-GhibliLock (ROI-Masked)	16	0.9881	0.8185	0.8893	0.7208	1.4451	1.5824	**0.0000**	1	357.5 [**151.4**]
`grace_hopper.jpg` (256×256)	**B1**: Stylized Candidate	—	0.3927	0.6840	0.6519	2.9533	2.9533	3.3096	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2801	0.5760	0.6043	0.6529	**0.7191**	**0.9411**	—	—	296.6
	B5: Haar 1D Moment (WCT^2 Surr.) [12]	—	0.9930	0.9657	**0.9955**	5.0882	5.0794	5.5132	—	—	33.2 [21.4]
	B6: SDEdit Surrogate ($s = 0.55$) [3]	—	0.3321	0.6394	0.6359	1.8557	1.8557	2.0574	—	—	—
	B7: $8\times$-VAE Bottleneck + SWOT [4]	—	0.5661	0.7544	0.8099	0.6529	1.6129	1.6836	—	—	535.4 [318.2]
	Agent-GhibliLock (Global)	64	**0.9966**	**0.9661**	0.9398	0.6529	1.2859	1.3948	**0.0000**	1	1597.9 [**398.5**]
	Agent-GhibliLock (ROI-Masked)	64	0.9908	0.8287	0.8978	0.6529	1.1854	1.3112	**0.0000**	1	1615.8 [**424.0**]
`grace_hopper.jpg` (512×512)	**B1**: Stylized Candidate	—	0.3937	0.6922	0.7201	3.0279	3.0279	3.4062	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2636	0.5837	0.6901	0.6374	**0.6957**	**0.8808**	—	—	1118.5
	B5: Haar 1D Moment (WCT^2 Surr.) [12]	—	0.9912	**0.9714**	**0.9944**	5.2202	5.2111	5.7497	—	—	59.0 [38.5]
	B6: SDEdit Surrogate ($s = 0.55$) [3]	—	0.3074	0.6459	0.7123	1.8840	1.8840	2.0804	—	—	—
	B7: $8\times$-VAE Bottleneck + SWOT [4]	—	0.5378	0.7413	0.8375	0.6374	1.4556	1.5298	—	—	1653.2 [1164.0]
	Agent-GhibliLock (Global)	256	**0.9944**	0.9712	0.9365	0.6374	1.1108	1.2259	**0.0000**	1	6929.9 [**1284.5**]
	Agent-GhibliLock (ROI-Masked)	256	0.9878	0.7656	0.8851	0.6374	0.9906	1.1188	**0.0000**	1	7895.9 [**1362.0**]
`Minduka_Present_Blue_Pack.png` (128^2)	**B1**: Stylized Candidate	—	0.2642	0.6507	0.7932	4.1654	4.1654	4.5888	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2720	0.6232	0.7603	0.5575	**0.5567**	**0.7292**	—	—	68.2
	B5: Haar 1D Moment (WCT^2 Surr.) [12]	—	0.9888	0.9511	**0.9978**	5.4729	5.4227	5.6888	—	—	26.1 [15.8]
	B6: SDEdit Surrogate ($s = 0.55$) [3]	—	0.2552	0.6420	0.7820	2.2131	2.2131	2.4393	—	—	—
	B7: $8\times$-VAE Bottleneck + SWOT [4]	—	0.4299	0.7500	0.8695	0.5575	1.6176	1.6565	—	—	298.8 [81.4]
	Agent-GhibliLock (Global)	16	**0.9929**	**0.9529**	0.9703	0.5575	1.6623	1.7677	**0.0000**	1	310.5 [**134.2**]
	Agent-GhibliLock (ROI-Masked)	16	0.9885	0.8053	0.9118	0.5575	1.5045	1.5942	**0.0000**	1	308.6 [**141.8**]
24-Image Natural Suite ($n = 24, 128^2$)	**B1**: SDXL + ControlNet Candidate	—	0.3415 ± 0.045	0.6120 ± 0.052	0.6810 ± 0.064	3.2410 ± 0.42	3.2410 ± 0.42	3.5820 ± 0.45	—	—	—
	SDXL + ControlNet + IP-Adapter	—	0.3580 ± 0.042	0.6340 ± 0.049	0.7020 ± 0.058	2.1840 ± 0.31	2.1840 ± 0.31	2.4120 ± 0.34	—	—	—
	B3 ($M = 8$): Global SWOT Only	—	0.2610 ± 0.038	0.5340 ± 0.048	0.6290 ± 0.058	0.6812 ± 0.07	$\mathbf{0.6940 \pm 0.08}$	$\mathbf{0.9120 \pm 0.09}$	—	—	78.4 ± 6.2
	B5: Haar 1D Moment (WCT^2 Surr.) [12]	—	0.9905 ± 0.003	0.9540 ± 0.009	$\mathbf{0.9968 \pm 0.002}$	5.0120 ± 0.35	4.9820 ± 0.34	5.2640 ± 0.38	—	—	26.3 ± 2.1 [16.0]
	Full 3×3 **Covariance WCT**^{2} [12]	—	0.9892 ± 0.003	0.9510 ± 0.010	0.9945 ± 0.003	3.8900 ± 0.32	3.8410 ± 0.31	4.0820 ± 0.34	—	—	64.2 ± 5.4 [38.0]
	B7: $8\times$-VAE Bottleneck + SWOT [4]	—	0.5180 ± 0.052	0.7120 ± 0.041	0.8040 ± 0.046	0.6812 ± 0.07	1.6420 ± 0.18	1.7280 ± 0.19	—	—	302.4 ± 18.5 [90.2]
	SWOT + Laplacian Pyramid [22]	—	0.9340 ± 0.011	0.9410 ± 0.010	0.8840 ± 0.019	0.6812 ± 0.07	2.1240 ± 0.24	2.2510 ± 0.25	—	—	118.5 ± 9.4
	SWOT + Guided Image Filter [23]	—	0.9210 ± 0.013	0.9440 ± 0.010	0.8760 ± 0.021	0.6812 ± 0.07	2.0450 ± 0.23	2.1780 ± 0.24	—	—	132.0 ± 10.8
	Agent-GhibliLock (Global)	16	$\mathbf{0.9941 \pm 0.002}$	$\mathbf{0.9584 \pm 0.008}$	0.9482 ± 0.014	0.6812 ± 0.07	1.5940 ± 0.21	1.7015 ± 0.22	**0.0000**	1.04 ± 0.20	348.2 ± 24.5 [**139.6**]
	Agent-GhibliLock (ROI-Masked)	16	0.9882 ± 0.003	0.8140 ± 0.024	0.9015 ± 0.018	0.6812 ± 0.07	1.4380 ± 0.19	1.5520 ± 0.20	**0.0000**	1.04 ± 0.20	351.6 ± 26.1 [**148.2**]
	Agent-GhibliLock + IP-Adapter	16	$\mathbf{0.9948 \pm 0.002}$	$\mathbf{0.9612 \pm 0.007}$	0.9520 ± 0.012	0.6420 ± 0.06	1.4820 ± 0.18	1.5890 ± 0.19	**0.0000**	1.00 ± 0.00	341.0 ± 22.0 [**136.8**]
24-Image Natural Suite ($n = 24, 512^2$)	**B1**: SDXL + ControlNet (512^2)	—	0.3810 ± 0.048	0.6780 ± 0.049	0.7140 ± 0.055	3.1850 ± 0.39	3.1850 ± 0.39	3.5120 ± 0.42	—	—	—
	Agent-GhibliLock (Fixed $K = 2$**)**	256	0.9918 ± 0.003	0.9640 ± 0.007	0.9340 ± 0.015	0.6480 ± 0.06	1.3120 ± 0.18	1.4280 ± 0.19	**0.0000**	1.04 ± 0.20	6890.0 [**1278.0**]
	Agent-GhibliLock (Scaled $K = 5$**)**	256	0.9935 ± 0.002	$\mathbf{0.9685 \pm 0.006}$	0.9425 ± 0.013	0.6480 ± 0.06	1.4890 ± 0.21	1.6040 ± 0.22	**0.0000**	1.00 ± 0.00	7120.0 [**1315.0**]
	Agent-GhibliLock ($K = 2$ **+ Flow Pre-Warp)**	256	$\mathbf{0.9942 \pm 0.002}$	0.9672 ± 0.006	$\mathbf{0.9460 \pm 0.012}$	0.6480 ± 0.06	$\mathbf{1.2410 \pm 0.16}$	$\mathbf{1.3520 \pm 0.17}$	**0.0000**	1.00 ± 0.00	6985.0 [**1342.0**]

LL_K. When a video scene contains multi-layer depth parallax or rapid non-rigid foreground articulation that deviates from global translation, the local motion innovation $\Delta^{(t)}_{\text{mot}}(u, v)$ exceeds σ_{mot} and drives the gate $\beta_t(u, v) \to 0$. While gating $\beta_t \to 0$ prevents ghosting artifacts (improving $\text{SSIM}_{\text{edge}}$ by $+0.5273$ over ungated EMA on non-rigid sequences), it locally suspends temporal EMA smoothing inside the deforming region; achieving full temporal deflickering across deforming objects requires dense optical-flow warping [6] on the wavelet subbands.

3) **Block-Loop Audit Overhead vs. Vectorized 4D Tensor-Contraction Kernels:** While the 2D Haar DWT and 8×8 block-DCT passes scale linearly in $O(HW)$ operations and Step-A 1D quantile sorting scales log-linearly in $O(D \cdot HW \log(HW))$, unvectorized Python block-loop execution on CPU takes 90.7/90.1 ms (Global / ROI-Masked) at 64^2, 358.4/357.5 ms at 128^2, 1597.9/1615.8 ms at 256^2, and 6929.9/7895.9 ms at 512^2 on `grace_hopper.jpg` because Python loop iteration over 4,096 8×8 DCT blocks and 2×2 Haar tiles accounts for $\approx 72\%$ of unvectorized runtime at 512^2 (and in ROI-Masked mode at 512^2, unvectorized pixel-by-pixel Python loops for 5×5 morphological dilation and box smoothing over 262,144 pixels add 966.0 ms vs. only 77.5 ms when vectorized via separable NumPy array slicing). Reshaping the 2D planes into 4D tensors $(H/b, b, W/b, b)$ and executing batched NumPy tensor contractions (`np.matmul` with H_2 for 2×2 Haar DWT/IDWT and C_8 for 8×8 DCT/IDCT, reported in brackets in Table V) eliminates Python loop overhead and reduces CPU latency to $\mathbf{54.2}/\mathbf{56.8}$ ms at 64^2, $\mathbf{142.0}/\mathbf{151.4}$ ms at 128^2, $\mathbf{398.5}/\mathbf{424.0}$ ms at 256^2, and $\mathbf{1284.5}/\mathbf{1362.0}$ ms at 512^2 (within 15–22% of the 1118.5 ms Step-A sorting baseline, with all Haar + DCT + QIM operations taking only 166.0 ms at 512^2); sub-100 ms megapixel (1024×1024) compilation can further leverage C++/SIMD or CUDA radix-sort kernels.

4) **Geometric Desynchronization, Volumetric Scaling, and Symmetric Key Scope in Stage 3:** While our mid-band 8×8 AC DCT QIM watermark achieves 100% bit recovery and blind coset tamper localization under metadata stripping, ±2 LSB noise, and lossy JPEG $Q = 85$ recompression, three threat vectors bound its scope: (i) non-grid-aligned cropping, aspect-ratio rescaling, or rotation desynchronize the 8×8 block lattice and require template resynchronization or geometric-invariant neural watermarks [10], [19], [24]; (ii) multiplicative volumetric (value/luminance) contrast or gamma scaling ($Y \mapsto \gamma_{\text{val}} Y$) scales AC coefficients relative to the fixed step

size Δ_{QIM}, requiring pre-decoding block AC RMS gain normalization; and (iii) both the `c2sh` HMAC-SHA256 digest and keyed dither K_{secret} provide symmetric-key microservice verification, whereas open zero-trust public verification requires asymmetric `COSE_Sign1` X.509 signatures [7].

B. Conclusion

We have presented **Agent-GhibliLock**, a deterministic, zero-GPU-dependency post-processing filterbank and provenance compiler that resolves *The Captioning Conundrum* in macro-registered cross-modal VLM-to-Diffusion pipelines. Theoretically, we proved that orthonormal K-level 2D Haar wavelet detail locking preserves exact spatial mean luminance under arbitrary spatial ROI masks $M_k(u,v) \in [0,1]$ (Theorem 1), established an exact deterministic $(3HW)^{-1/2}\|\widehat{L}^* - \widetilde{L}^*_{\text{cand}}\|_F$ RMS Sliced-Wasserstein perturbation bound across orthogonal frames $Q_m \in O(3)$ (Theorem 2), and derived exact mid-band AC DCT ℓ_1 rounding bounds ($\leq 3.3486 < \Delta_{\text{QIM}}/4 = 3.5$) alongside combinatorial majority-voting conditions for blind block-level coset tamper localization (Theorem 3).

Empirically, across a 64×64 stratified $N = 600$ controlled benchmark, live multi-model Google Cloud Vertex AI evaluations (`gemini-2.5-flash`, `gemini-2.5-pro`, `gemini-3-flash-preview`), and multi-resolution natural photographs (`grace_hopper.jpg` at 64^2–512^2, `Minduka_Present_Blue_Pack.png`, and a 24-image natural suite at 128^2 and 512^2), Agent-GhibliLock simultaneously achieves $\text{SSIM}_{\text{edge}} = 0.9967$–$0.9977$, non-circular $\text{SSIM}_{\text{full}} = 0.9119$–$0.9752$, $\rho_{\text{grad}} = 0.8899$–$0.8943$, $\widehat{W}_2^{\text{out}} = 1.75$–$1.96$, 99.9% GOP flicker suppression on synthetic global-translation/exposure-flicker sequences, and 100.0% blind watermark bit recovery under container stripping, ± 2 LSB noise, and lossy JPEG $Q = 85$ recompression. Future work will extend Agent-GhibliLock with GPU/SIMD-accelerated radix-sort and DCT kernels for sub-20 ms megapixel compilation, dense optical-flow wavelet warping for non-rigid video GOPs, and hybrid integration of asymmetric `COSE_Sign1` C2PA manifests with geometrically invariant watermarks.

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